

Phase 1: Planning & Design

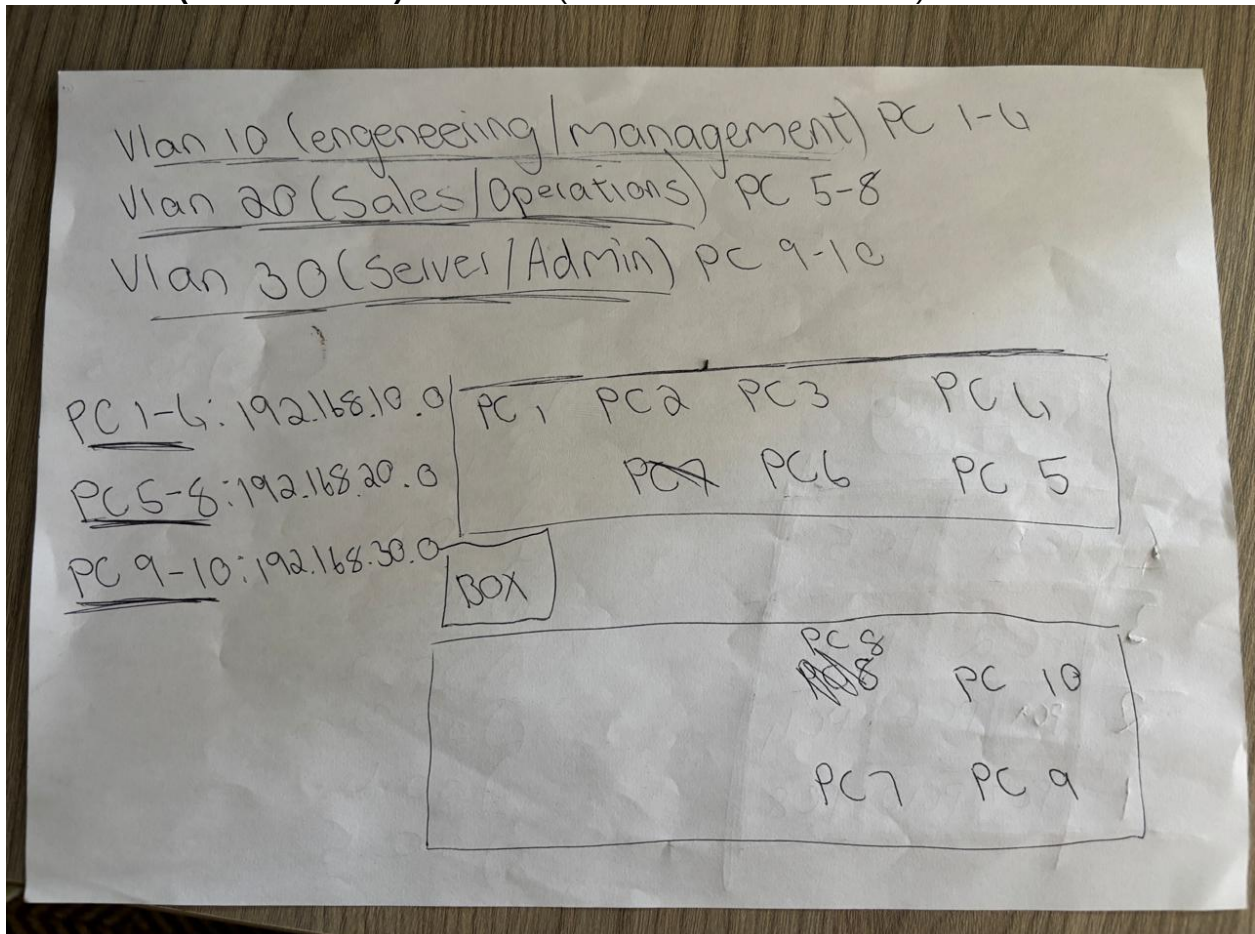
1. Physical Topology

- **The Single Switch Cabinet (Rack):**
 - **U1–U4 (Top):** 4 × D-Link Patch Panels (Grouped together at the top to aggregate incoming PC lines safely).
 - **U5–U6 (Middle):** 2 × Cisco Catalyst 3560 Switches (Placed directly under the patch panels for short, clean patch cord routing).
 - **U7–U8 (Bottom):** 2 × Cisco ISR 4331 Routers (Placed at the bottom due to heavier weight and separate power requirements).
- **The Device Connections:** 10 PCs sit externally. Their solid-core cables run directly into the back of the 4 patch panels inside the cabinet.

2. Logical Topology & VLAN Infrastructure

To accommodate 10 PCs across 2 switches and 2 routers, the network should be segmented logically into departmental VLANs to demonstrate routing capability.

- **VLAN 10 (Engineering / Management):** PCs 1–4 (Connected to Switch 1)
- **VLAN 20 (Sales / Operations):** PCs 5–8 (Connected to Switch 2)
- **VLAN 30 (Server / Admin):** PCs 9–10 (Connected to Switch 1 or 2)



3.

- **Native/Management VLAN 99:** For Switches and Routers IP addresses.

4. IP Address Planning

Using a private Class C subnet space (**192.168.0.0/24**), divided using VLSM:

- **VLAN 10 Subnet:** 192.168.10.0/26 (Gateway: .1)
- **VLAN 20 Subnet:** 192.168.20.0/26 (Gateway: .1)
- **VLAN 30 Subnet:** 192.168.30.0/26 (Gateway: .1)
- **Management VLAN 99:** 192.168.99.0/24
- **Router-to-Router Link:** 10.0.0.0/30 (R1: .1, R2: .2)

Phase 2: Hardware Installation & Cabling

1. **Install Switch Cabinet:** Securely mount the rack enclosure. Ensure grounding cables are bonded to the cabinet chassis.
2. **Install Patch Panels:** Screw all 4 D-Link patch panels into the top rack units (U1–U4).
3. **Terminate Internal Cable Runs:**
 - Route the raw CAT6 cables coming from the 10 PC stations into the cabinet.
 - Pull them to the *back* side of the patch panels.
 - Use the **insertion (punch-down) tool** to map the wires directly to the **T568B** color-coded blocks.
 - *Note: Because outlets/faceplates were removed, the PC side of the cables must be*

punched down into loose CAT6 keystone jacks sitting at each desk.

4. **Test Cabling:** Use a hardware cable continuity tester on the punched-down lines to ensure no split pairs or shorts exist before powering up electronics.

Phase 3: Device Initialization & Layer 2 Config

1. **Physical Console Access:** Connect a PC running **PuTTY** to the Cisco devices using the provided **Console cables** (RJ45-to-DB9 or USB-to-RJ45) plugged into the device's Console port.
2. **Basic Configuration (Switches & Routers):**
Router(config)# hostname R1
R1(config)# enable secret *YourStrongPassword*
R1(config)# line vty 0 4
R1(config-line)# password *YourVtyPassword*
R1(config-line)# login
3. **Switch Management Interfaces:**
SW1(config)# vlan 99
SW1(config-vlan)# name Management
SW1(config)# interface vlan 99
SW1(config-if)# ip address 192.168.99.10 255.255.255.0
SW1(config-if)# no shutdown
4. **Configure VLANs & Access Ports:**
SW1(config)# vlan 10
SW1(config-vlan)# name Engineering
SW1(config)# interface range fa0/1 - 4
SW1(config-if-range)# switchport mode access
SW1(config-if-range)# switchport access vlan 10
5. **Trunking between Switches:** Configure Inter-Switch Links (ISL/802.1Q) to allow VLAN traffic to traverse both Catalyst 3560s.

Phase 4: Layer 3 Configuration & Routing

1. **Configure Router Interfaces:** Assign IPs to the physical interfaces connecting the two Cisco ISR4331 routers together.
2. **Inter-VLAN Routing (Router-on-a-Stick):** Connect a trunk port from Switch 1 to Router 1's GigabitEthernet interface. Create sub-interfaces on the router to act as default gateways for the PCs:
R1(config)# interface gi0/0/0.10
R1(config-subif)# encapsulation dot1Q 10
R1(config-subif)# ip address 192.168.10.1 255.255.255.192

3. **Configure Routing Protocols:** Set up static routing or OSPF between Router 1 and Router 2 so networks on Switch 1 can communicate with networks on Switch 2.
4. **Verify Routing Tables:** Run show ip route via PuTTY to ensure all subnets are visible.

Phase 5: Network Printers (Skipped / Simulated)

- **Status: Omitted per instructions.**
- *Alternative Action:* If the instructor requires verification of this phase despite missing hardware, assign an unused IP address (e.g., 192.168.30.50) to a loopback interface on a switch or router to act as a "Virtual Printer," then verify that all VLANs can successfully ping it.

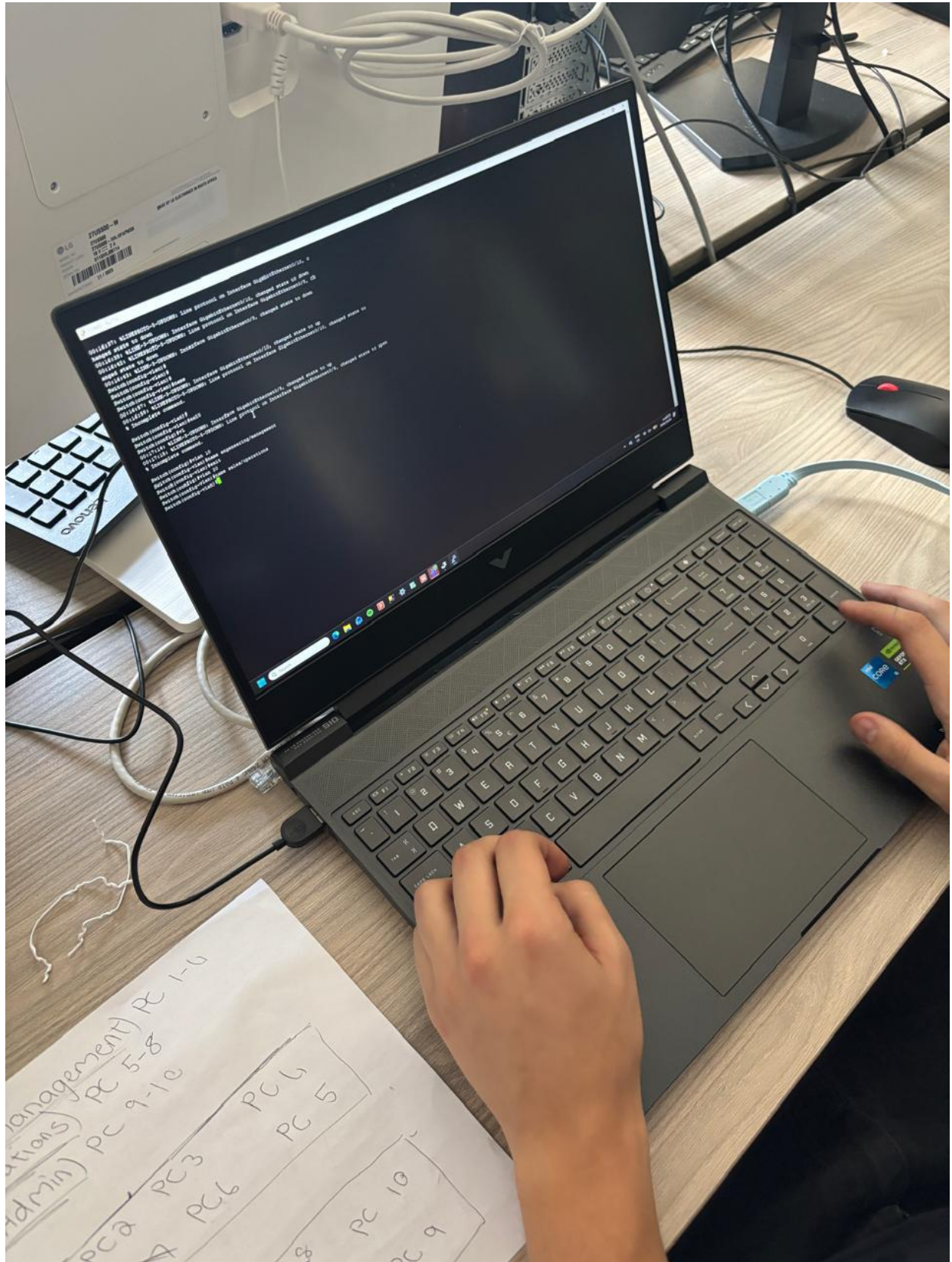
Phase 6: Testing & Troubleshooting

1. **Verify Connectivity:** Ping from a PC in VLAN 10 to a PC in VLAN 20.
2. **Verify VLAN Functionality:** Use show vlan brief on the switches to confirm ports are grouped correctly.
3. **Resolve Instructor Faults:** Common simulated faults to look out for via PuTTY command line:
 - **Wrong VLAN Assignment:** Fixed using switchport access vlan X.
 - **Native VLAN Mismatch:** Fixed by matching native IDs on both ends of a trunk link.
 - **Subnet Mask Mismatch:** Check PC IP configurations vs the Router sub-interface gateway.

Phase 7: Documentation

Produce a final network book containing:

- Completed Visio/Packet Tracer **Logical Diagram**.
- High-level **Cabinet Rack Layout map** (Physical topology).
- Master **IP Addressing Matrix table** (listing every PC, Switch, Router interface, and functional gateway).
- Text file backups of the running configurations (show running-config) saved directly from the PuTTY session logs.



```
COM3 - PuTTY
Switch(config-vlan)#name sales/operations: is down
Switch(config-vlan)#exit: does not set protocol on interface GigabitEthernet0/10, s
Switch(config)#vlan 30: down, line protocol is down
Switch(config-vlan)#name server/admin: down, line protocol is down
Switch(config-vlan)#exit: down, line protocol is down
Switch(config)#no list: is not set
Switch(config)#exit: is down, line protocol is down
Switch#show ip: is not set
00:18:13: SYS-5-CONFIG_I: Configured from console by consolep in
Vlan1 is up, line protocol is up:
Internet protocol processing disabled: is down
GigabitEthernet0/1: is down, line protocol is down
GigabitEthernet0/2: is up, line protocol is up
GigabitEthernet0/3: is up, line protocol is up
GigabitEthernet0/4: is up, line protocol is up
GigabitEthernet0/5: is up, line protocol is up
GigabitEthernet0/6: is up, line protocol is up
GigabitEthernet0/7: is up, line protocol is up
GigabitEthernet0/8: is up, line protocol is up
GigabitEthernet0/9: is down, line protocol is down
GigabitEthernet0/10: is up, line protocol is up
GigabitEthernet0/11: is down, line protocol is down
GigabitEthernet0/12: is down, line protocol is down
GigabitEthernet0/13: is down, line protocol is down
GigabitEthernet0/14: is down, line protocol is down
GigabitEthernet0/15: is down, line protocol is down
GigabitEthernet0/16: is down, line protocol is down
GigabitEthernet0/17: is down, line protocol is down
GigabitEthernet0/18: is down, line protocol is down
GigabitEthernet0/19: is down, line protocol is down
GigabitEthernet0/20: is down, line protocol is down
GigabitEthernet0/21: is down, line protocol is down
GigabitEthernet0/22: is down, line protocol is down
GigabitEthernet0/23: is down, line protocol is down
GigabitEthernet0/24: is down, line protocol is down
GigabitEthernet0/25: is down, line protocol is down
GigabitEthernet0/26: is down, line protocol is down
```

Switch#
Switch#
Switch#
Switch#
Switch#

Switch#show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
Vlan1	unassigned	YES	unset	up	up
GigabitEthernet0/1	unassigned	YES	unset	down	down
GigabitEthernet0/2	unassigned	YES	unset	up	up
GigabitEthernet0/3	unassigned	YES	unset	up	up
GigabitEthernet0/4	unassigned	YES	unset	up	up
GigabitEthernet0/5	unassigned	YES	unset	up	up
GigabitEthernet0/6	unassigned	YES	unset	up	down
GigabitEthernet0/7	unassigned	YES	unset	up	up
GigabitEthernet0/8	unassigned	YES	unset	down	down
GigabitEthernet0/9	unassigned	YES	unset	up	down
GigabitEthernet0/10	unassigned	YES	unset	down	down
GigabitEthernet0/11	unassigned	YES	unset	down	down
GigabitEthernet0/12	unassigned	YES	unset	down	down
GigabitEthernet0/13	unassigned	YES	unset	down	down
GigabitEthernet0/14	unassigned	YES	unset	down	down
GigabitEthernet0/15	unassigned	YES	unset	down	down
GigabitEthernet0/16	unassigned	YES	unset	down	down
GigabitEthernet0/17	unassigned	YES	unset	down	down
GigabitEthernet0/18	unassigned	YES	unset	down	down
GigabitEthernet0/19	unassigned	YES	unset	down	down
GigabitEthernet0/20	unassigned	YES	unset	down	down
GigabitEthernet0/21	unassigned	YES	unset	down	down
GigabitEthernet0/22	unassigned	YES	unset	down	down
GigabitEthernet0/23	unassigned	YES	unset	down	down
GigabitEthernet0/24	unassigned	YES	unset	down	down
GigabitEthernet0/25	unassigned	YES	unset	down	down
GigabitEthernet0/26	unassigned	YES	unset	down	down
GigabitEthernet0/27	unassigned	YES	unset	down	down
GigabitEthernet0/28	unassigned	YES	unset	down	down

Switch#
Switch#
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Switch#
Switch#
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Switch#

Switch#show vlan

VLAN Name	Status	Ports
1 default	active	G10/1, G10/2, G10/3, G10/4, G10/5, G10/6, G10/7, G10/8, G10/9, G10/10, G10/11, G10/12, G10/13, G10/14, G10/15, G10/16, G10/17, G10/18, G10/19, G10/20, G10/21, G10/22, G10/23, G10/24, G10/25, G10/26, G10/27, G10/28

- 10 engineering/management
- 20 sales/operations
- 30 server/admin
- 1002 fddi-default

```

GigabitEthernet0/13 unassigned YES unset down down
GigabitEthernet0/14 unassigned YES unset down down
GigabitEthernet0/15 unassigned YES unset down down
GigabitEthernet0/16 unassigned YES unset down down
GigabitEthernet0/17 unassigned YES unset down down
GigabitEthernet0/18 unassigned YES unset down down
GigabitEthernet0/19 unassigned YES unset down down
GigabitEthernet0/20 unassigned YES unset down down
GigabitEthernet0/21 unassigned YES unset down down
GigabitEthernet0/22 unassigned YES unset down down
GigabitEthernet0/23 unassigned YES unset down down
GigabitEthernet0/24 unassigned YES unset down down
GigabitEthernet0/25 unassigned YES unset down down
GigabitEthernet0/26 unassigned YES unset down down
GigabitEthernet0/27 unassigned YES unset down down
GigabitEthernet0/28 unassigned YES unset down down

```

Switch#

Switch#

Switch#

Switch#

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Switch#

Switch#

Switch#show vlan

VLAN Name	Status	Ports
1 default	active	G10/1, G10/2, G10/3, G10/4 G10/5, G10/6, G10/7, G10/8 G10/9, G10/10, G10/11, G10/12 G10/13, G10/14, G10/15, G10/16 G10/17, G10/18, G10/19, G10/20 G10/21, G10/22, G10/23, G10/24 G10/25, G10/26, G10/27, G10/28
10 engineering/management	active	
20 sales/operations	active	
30 server/admin	act/unsup	
1002 fddi-default	act/unsup	
1003 token-ring-default	act/unsup	
1004 fddinet-default	act/unsup	
1005 trnet-default	act/unsup	

VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2
1	enet	100001	1500	-	-	-	-	-	0	0
10	enet	100010	1500	-	-	-	-	-	0	0
20	enet	100020	1500	-	-	-	-	-	0	0

--More--



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```

Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
00:19:39: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/4, changed state to down
00:19:40: %LINK-3-UPDOWN: Interface GigabitEthernet0/4, changed state to down
00:19:43: %LINK-3-UPDOWN: Interface GigabitEthernet0/4, changed state to down
00:19:44: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/4, changed state to up
Switch#
Switch#
Switch#
Switch#interface vlan 10
Switch#
^
% Invalid input detected at '^' marker.

Switch#confi ter
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface vln 10
Switch#
^
% Invalid input detected at '^' marker.

Switch(config)#INTERFACE VLAN 10
Switch(config-if)#
00:20:30: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan10, changed state to down
Switch(config-if)#IP ADDRESS 192.168.10.0 255.255.255.0
Bad mask /24 for address 192.168.10.0
Switch(config-if)#
00:21:13: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/5, changed state to down
00:21:14: %LINK-3-UPDOWN: Interface GigabitEthernet0/5, changed state to down
Switch(config-if)#IP ADDRESS 192.168.10.1 255.
00:21:28: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to down
00:21:29: %LINK-3-UPDOWN: Interface GigabitEthernet0/2, changed state to down255
% Incomplete command.

Switch(config-if)#IP ADDRES 1
00:21:33: %LINK-3-UPDOWN: Interface GigabitEthernet0/2, changed state to up
00:21:33: %LINK-3-UPDOWN: Interface GigabitEthernet0/5, changed state to up
00:21:34: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up92.
00:21:34: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/5, changed state to up
00:21:34: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/5, changed state to up
% Incomplete command.

Switch(config-if)#IP ADDRESS 192.168.10.1 25
00:21:44: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/4, changed state to down
00:21:45: %LINK-3-UPDOWN: Interface GigabitEthernet0/4, changed state to down5.IP ADDRESS 192.168.10.1 25
Switch#
^
% Invalid input detected at '^' marker.

Switch(config-if)#IP ADDRESS 192.168.10.1 25
% Incomplete command.

Switch(config-if)#
00:21:49: %LINK-3-UPDOWN: Interface GigabitEthernet0/4, changed state to up
00:21:50: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/4, changed state to up
Switch(config-if)#IP ADDRESS 192.168.10.1 255.255.255.0
Switch(config-if)#EXIT
Switch(config)#INTERFACE VLAN 20
Switch(config-if)#
00:22:40: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan20, changed state to down

```

access-list
alias
archive
arp
banner
boot
buffers
cdp
cef
class-map
clock
cluster
cns
configuration
control-plane
default
default-value
define
diagnostic
do

Add an access list entry
Create command alias
Archive the configuration
Set a static ARP entry
Define a login banner
Boot Commands
Adjust system buffer pool parameters
Global CDP configuration subcommands
Cisco Express Forwarding
Configure QoS Class Map
Configure time-of-day clock
Cluster configuration commands
CNS agents
Configuration access
Configure control plane services
Set a command to its defaults
Default character-bits values
Interface range macro definition
Configure diagnostic information
To run exec commands in config mode

Switch(config)#define?
define

Switch(config)#define ?
interface-range interface-range

Switch(config)#define interface-range f0/1-4
? Incomplete command.

Switch(config)#define interface f0/1
? Incomplete command.

Switch(config)#interface-range
? Invalid input detected at ' ' marker.

Switch(config)#interface ?
Async Async interface
SVI Bridge-Group Virtual Interface
CTunnel CTunnel interface
Dialer Dialer interface
Filter Filter interface
Filtergroup Filter Group interface
GigabitEthernet GigabitEthernet IEEE 802.3z
Group-Async Async Group interface
lex Lex interface
Loopback Loopback interface
Null Null interface
Port-channel Ethernet interface
Portgroup Portgroup interface
Pos-channel POS Channel of interfaces
Tunnel Tunnel interface
Vif Vif interface
Virtual-Template PGM Multicast Host interface
Virtual-TokenRing Virtual Template interface
Vlan Catalyst Vlans
fcpa Fiber Channel
range interface range command

Switch(config)#interface g0/1
Switch(config-if)#



```
Configure...
Set a command to...
Default character-bits value
Interface range macro definition
Interface diagnostic information
Configure diagnostic information
To run exec commands in config mode

line ?
  interface-range
  define interface-range f0/1-4
  command.
  (ig)#define interface f0/1
  te command.
  (config)#interface-range
    ^
    Invalid input detected at '^' marker.

Switch(config)#interface ?
  Async
  Async interface
  Bridge-Group Virtual Interface
  CTunnel
  CTunnel interface
  Dialer
  Dialer interface
  Filter
  Filter Group interface
  Filter Group IEEE 802.3z
  GigabitEthernet
  GigabitEthernet interface
  Loopback
  Loopback interface
  Lex
  Lex interface
  Loopback interface
  Null
  Null interface
  Portgroup
  Portgroup interface
  Portgroup Channel of interfaces
  Portgroup Channel of interfaces
  Tunnel
  Tunnel interface
  Tunnel Multicast Host interface
  Virtual-Template
  Virtual-Template interface
  Virtual-TokenRing
  Virtual-TokenRing interface
  Vlan
  Vlan interface
  Vlan range
  Vlan range command
  Vlan range command

Switch(config)#interface g0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface g0/1
```



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Interface ?
Access-range
Accessed at *** marker.

- Async interface
- Bridge-Group Virtual Interface
- CTunnel interface
- Dialer interface
- Filter interface
- Filter Group interface
- GigabitEthernet IEEE 802.3z
- Async Group interface
- Lex interface
- Loopback interface
- Null interface
- Ethernet Channel of interfaces
- Portgroup interface
- POS Channel of interfaces
- Tunnel interface
- PGM Multicast Host interface
- Virtual Template interface
- Virtual TokenRing
- Catalyst Vlans
- Fiber Channel
- Interface range command

Switch(config)#interface g0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface g0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#

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Input Detected at '^' marker.

config)#interface ?

Async interface
Bridge-Group Virtual Interface
CTunnel interface
Dialer interface
Filter interface
Filter Group interface
GigabitEthernet IEEE 802.3z
Async Group interface
Lex interface
Loopback interface
Null interface
Ethernet Channel of interfaces
Portgroup interface
POS Channel of interfaces
Tunnel interface
PGM Multicast Host interface
Virtual Template interface
Virtual TokenRing
Catalyst Vlan
Fiber Channel
interface range command
range

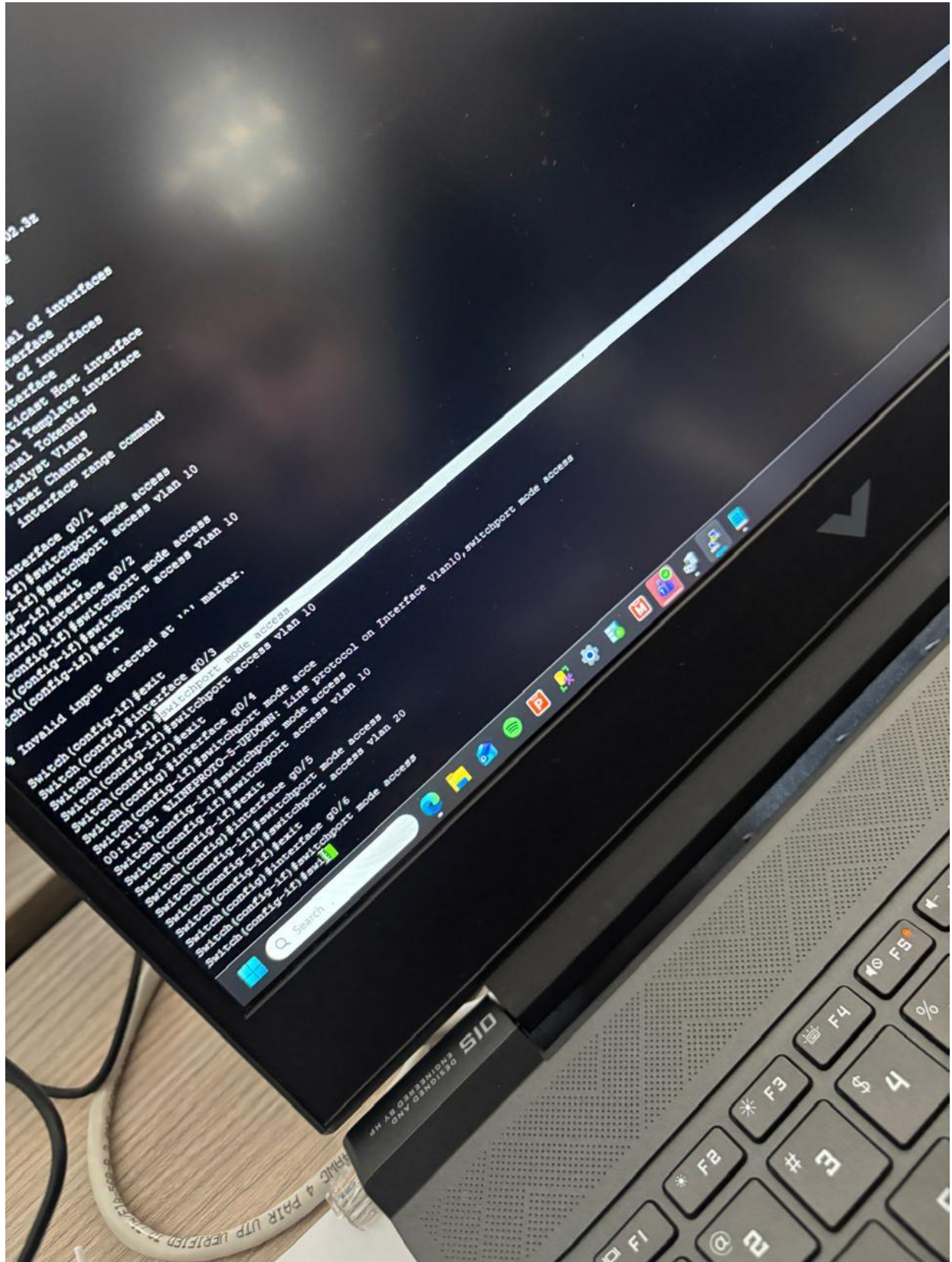
Switch(config)#interface g0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface g0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config-if)#exit
Switch(config-if)#exit

Invalid input detected at '^' marker.

Switch(config-if)#exit
Switch(config)#interface g0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface g0/4
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
00:31:35: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan10, changed state to

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```

Async
BVI
CTunnel
Dialer
Filter
Filtergroup
GigabitEthernet
Group-Async
Lex
Loopback
Null
Port-channel
Portgroup
Pos-channel
Tunnel
Vif
Virtual-Template
Virtual-TokenRing
Vlan
fcpa
range

Async interface
Bridge-Group Virtual Interface
CTunnel interface
Dialer interface
Filter interface
Filter Group interface
GigabitEthernet IEEE 802.3z
Async Group interface
Lex interface
Loopback interface
Null interface
Ethernet Channel of interfaces
Portgroup interface
POS Channel of interfaces
Tunnel interface
PGM Multicast Host interface
Virtual Template interface
Virtual TokenRing
Catalyst Vlan
Fiber Channel
interface range command

```

```

Switch(config)#interface g0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface g0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
^
% Invalid input detected at '^' marker.

Switch(config-if)#exit
Switch(config)#interface g0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface g0/4
Switch(config-if)#switchport mode access
00:31:35: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan10, switchport mode access
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface g0/5
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface g0/6
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface g0/7
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface g0/8
00:32:43: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan20, changed state to up
Switch(config)#interface g0/9
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#

```



```
Switch(config-if)#switchport mode access
Switch(config-if)#exit
^
% Invalid input detected at '^' marker.

Switch(config-if)#exit
Switch(config)#interface g0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface g0/4
Switch(config-if)#switchport mode access
00:31:35: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan10, switchport mode access
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface g0/5
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface g0/6
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface g0/7
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#
00:32:43: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan20, changed state to up
Switch(config)#interface g0/8
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface g0/9
^
% Invalid input detected at '^' marker.

Switch(config)#interface g0/9
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#exit
Switch(config)#interface g0/10
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#
00:33:50: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to down
```



Search



```

Switch(config)#interface g0/5
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface g0/6
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface g0/7
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
00:32:43: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan20, changed state to up
Switch(config)#interface g0/8
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface g0/9
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#exit
Switch(config)#interface g0/10
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#exit
00:33:15: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to down
Switch(config)#show vlan brief
Switch(config)#
% Invalid input detected at '^' marker.

Switch(config)#exit
Switch#show
00:34:02: %SYS-5-CONFIG_I: Configured from console by consolevian

VLAN Name                Status    Ports
-----
1    default                active    Gi0/11, Gi0/12, Gi0/13, Gi0/14,
                                           Gi0/15, Gi0/16, Gi0/17, Gi0/18,
                                           Gi0/19, Gi0/20, Gi0/21, Gi0/22,
                                           Gi0/23, Gi0/24, Gi0/25, Gi0/26,
                                           Gi0/27, Gi0/28
10   engineering/management active    Gi0/1, Gi0/2, Gi0/3, Gi0/4
20   sales/operations       active    Gi0/5, Gi0/6, Gi0/7, Gi0/8
30   server-admin            active    Gi0/9, Gi0/10
1002 FGD1-default            act/upnp
1003 token-ring-default    act/upnp
1004 dot1q-tunnel-default act/upnp
1005 vrrp1-default          act/upnp

VLAN Type  SAID             MTU    Parent Ringid BridgeId  Stp    BridgeMode Transl Transl
-----
1    enet  100001          1500    -    -    -    -    0        0
10   enet  100010          1500    -    -    -    -    0        0
20   enet  100020          1500    -    -    -    -    0        0
30   enet  100030          1500    -    -    -    -    0        0
1002 enet  101002          1500    -    -    -    -    0        0
--More--

```

```

Switch(config)#exit
Switch(config)#interface g0/7
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#
00:32:43: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan20, changed state to up
Switch(config)#interface g0/8
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface g0/9
Switch(config-if)#
^
% Invalid input detected at '^' marker.

```

```

Switch(config)#interface g0/9
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#exit
Switch(config)#interface g0/10
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#
00:33:50: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to down
Switch(config-if)#exit
Switch(config)#show vlan brief
^
% Invalid input detected at '^' marker.

```

```

Switch(config)#exit
Switch#show
00:34:02: %SYS-5-CONFIG_I: Configured from console by consolevlan

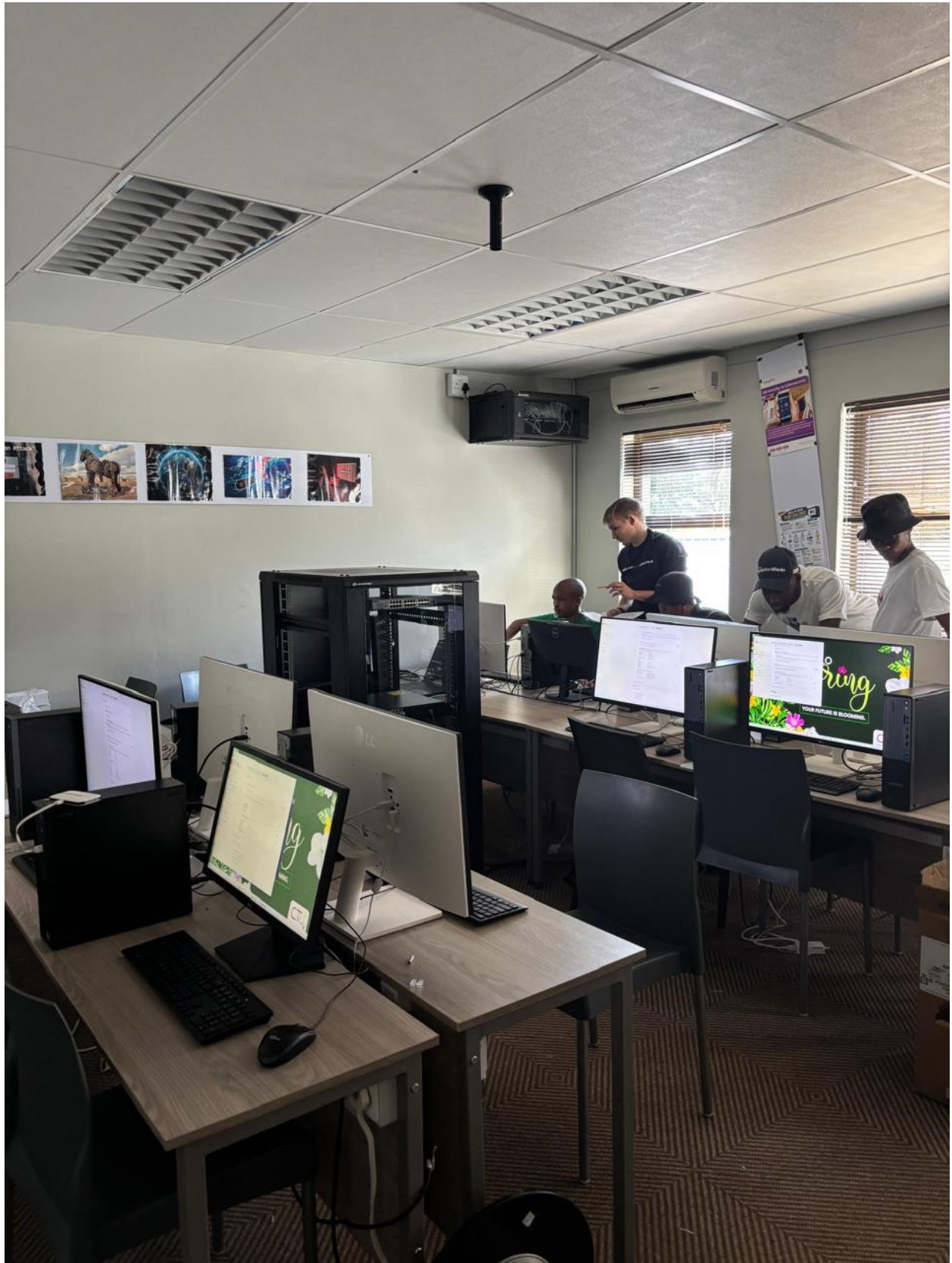
```

VLAN	Name	Status	Ports
1	default	active	G10/11, G10/12, G10/13, G10/14 G10/15, G10/16, G10/17, G10/18 G10/19, G10/20, G10/21, G10/22 G10/23, G10/24, G10/25, G10/26 G10/27, G10/28
10	engineering/management	active	G10/1, G10/2, G10/3, G10/4
20	sales/operations	active	G10/5, G10/6, G10/7, G10/8
30	server/admin	active	G10/9, G10/10
1002	fdi-default	act/unsup	
1003	token-ring-default	act/unsup	
1004	fdinet-default	act/unsup	
1005	trinet-default	act/unsup	

VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2
1	enet	100001	1500	-	-	-	-	-	0	0
10	enet	100010	1500	-	-	-	-	-	0	0
20	enet	100020	1500	-	-	-	-	-	0	0
30	enet	100030	1500	-	-	-	-	-	0	0
1002	fdi	101002	1500	-	-	-	-	-	0	0

--More--





Metered connection

Some apps might work differently to reduce data usage when you're connected to this network

Set a data limit to help control data usage on this network

Random hardware addresses

Help protect your privacy by making it harder for people to track your device location when you connect to this network. This feature takes effect the next time you connect to this network.

IP assignment:	Manual
IPv4 address:	192.168.20.4
IPv4 mask:	255.255.255.0
DNS server assignment:	Automatic (DHCP)
SSID:	CTU-Staff
Protocol:	Wi-Fi 6 (802.11ax)
Security type:	WPA2-Personal
Manufacturer:	Intel Corporation
Description:	Intel(R) Wi-Fi 6 AX201 160MHz
Driver version:	24.50.0.4
Network band (channel):	5 GHz (128)
Aggregated link speed (Receive/Transmit):	2402/2402 (Mbps)
Link-local IPv6 address:	fe80::b1d6:9582:68bb:356e%6
IPv6 DNS servers:	fec0:0:0:ffff::1%1 (Unencrypted) fec0:0:0:ffff::2%1 (Unencrypted) fec0:0:0:ffff::3%1 (Unencrypted)
IPv4 address:	192.168.20.4
Physical address (MAC):	38:18:68:62:34:91

Advanced Wi-Fi network properties

Related support



Search



Find a setting

Network & internet > Ethernet



Ethernet
Not connected

Authentication settings

Metered connection

Some apps might work differently to reduce data usage when you're connected to this network

[Set a data limit to help control data usage on this network](#)

IP assignment:	Manual
IPv4 address:	192.169.30.2
IPv4 mask:	255.255.255.0
DNS server assignment:	Automatic (DHCP)
Manufacturer:	Realtek
Description:	Realtek PCIe GbE Family Controller
Driver version:	1168.28.1224.2025
Physical address (MAC):	CC:96:E5:09:E5:73

Related support

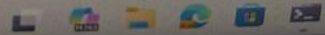


Help with Ethernet

- [Updating network adapter or driver](#)
- [Fixing Ethernet connection issues](#)
- [Run an internet speed test](#)
- [Finding my IP address](#)



Search



Command Prompt

```
TEMP.AzureAD.002>ipconfig
```

IP Configuration

LAN adapter Local Area Connection* 1:

Media State : Media disconnected
Connection-specific DNS Suffix . :

Local Area Connection* 2:

Media State : Media disconnected
Connection-specific DNS Suffix . :

Wireless LAN adapter Wi-Fi:

Media State : Media disconnected
Connection-specific DNS Suffix . :

Ethernet adapter Ethernet:

Connection-specific DNS Suffix . :
Link-local IPv6 Address : fe80::b2f1:a954:55d9:2878%16
IPv4 Address. : 192.168.30.3
Subnet Mask : 255.255.255.0
Default Gateway :

```
C:\Users\TEMP.AzureAD.002>
```

Network & Internet > Ethernet

Unidentified network

No internet

Authentication settings

Metered connection

Some apps might work differently to reduce data usage when you're connected to this network

[Set a data limit to help control data usage on this network](#)

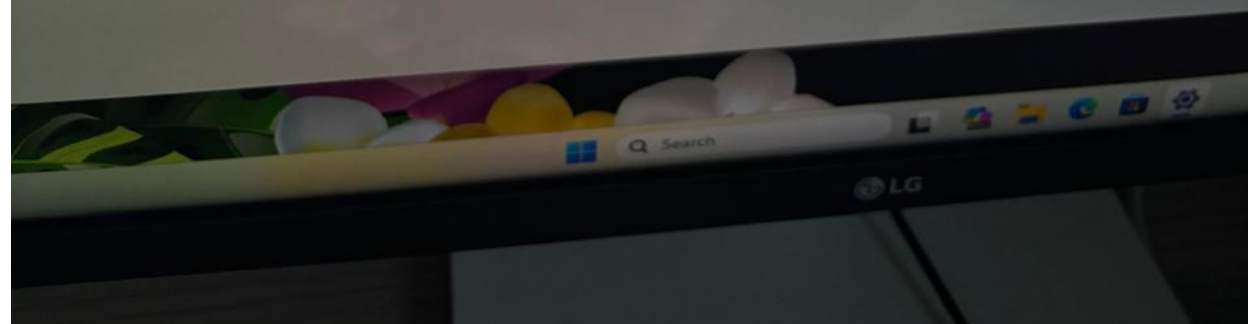
IP assignment:	Manual
IPv4 address:	192.168.20.5
IPv4 mask:	255.255.255.0

DNS server assignment:	Automatic (DHCP)
------------------------	------------------

Aggregated link speed (Receive/Transmit):	10/10 (Mbps)
---	--------------

Link-local IPv6 address:	fe80::4d3b:633c:7812:e729%4
IPv6 DNS servers:	fec0:0:0:ffff::1%1 (Unencrypted)
	fec0:0:0:ffff::2%1 (Unencrypted)
	fec0:0:0:ffff::3%1 (Unencrypted)

IPv4 address:	192.168.20.5
Manufacturer:	Realtek
Description:	Realtek PCIe GbE Family Controller
Driver version:	1168.28.1224.2025
Physical address (MAC):	CC:96:E5:09:EC:43



etered connection

me apps might work differently to reduce data usage when you're connected to this netv

et a data limit to help control data usage on this network

Random hardware addresses

Help protect your privacy by making it harder for people to track your device location when y
takes effect the next time you connect to this network.

IP assignment:	Manual
IPv4 address:	192.168.20.6
IPv4 mask:	255.255.255.0

DNS server assignment:	Automatic (DHCP)
------------------------	------------------

SSID:	
Protocol:	Any radio type
Security type:	Unknown
Manufacturer:	Intel Corporation
Description:	Intel(R) Wi-Fi 6 AX201 160MHz
Driver version:	24.50.0.4

Aggregated link speed (Receive/ Transmit):	2402/2402 (Mbps)
Link-local IPv6 address:	fe80::354b:db7e:c8f4:f925%15
IPv6 DNS servers:	fec0:0:0:ffff:1%1 (Unencrypted) fec0:0:0:ffff:2%1 (Unencrypted) fec0:0:0:ffff:3%1 (Unencrypted)
Physical address (MAC):	64 4A 7D 58 D7 04

Advanced Wi-Fi network properties

Related support

Connect automatically when in range

Work profile type

Public network (Recommended)

Your device is not discoverable on the network. Use this in most cases—when connected to a network in a public place.

Private network

Your device is discoverable on the network. Select this if you need file sharing or use apps that communicate with other devices on the network. You should know and trust the people and devices on the network.

Configure firewall and security settings

Metered connection

Some apps might work differently to reduce data usage when you're connected to this network.

Set a data limit to help control data usage on this network

Random hardware addresses

Help protect your privacy by making it harder for people to track your device location when you connect to this network. The setting takes effect the next time you connect to this network.

IP assignment:	Manual
IPv4 address:	192.168.20.5
IPv4 mask:	255.255.255.0
DNS server assignment:	Automatic (DHCP)

Advanced Wi-Fi network properties

Related support



Help from the web

ation

hernet:

```
Specific DNS Suffix . . :  
Address . . . . . : fe80::84c9:b22d:ffba:d80b%2  
 . . . . . : 192.168.20.5  
 . . . . . : 255.255.255.0  
y . . . . . :
```

apter Local Area Connection* 1:

```
 . . . . . : Media disconnected  
Specific DNS Suffix . . :
```

apter Local Area Connection* 2:

```
 . . . . . : Media disconnected  
Specific DNS Suffix . . :
```

apter Wi-Fi:

```
-specific DNS Suffix . . : ctutrainig.co.za  
IPv6 Address . . . . . : fe80::8989:71b4:9cc9:3aee%12  
ss. . . . . : 172.16.3.170  
sk . . . . . : 255.255.255.0  
ateway . . . . . : 172.16.3.1
```



tered connection

ne apps might work differently to reduce data usage when you're connected to this network

a data limit to help control data usage on this network

Assignment:	Manual
IPv4 address:	12.168.10.4
Subnet mask:	255.255.255.0

DNS server assignment:	Automatic (DHCP)
------------------------	------------------

Aggregated link speed (Receive/transmit):	10/10 (Mbps)
---	--------------

Link-local IPv6 address:	fe80::12cc:216b:8eb0:4408%11
--------------------------	------------------------------

IPv6 DNS servers:	fec0:0:0:ffff::1%1 (Unencrypted)
	fec0:0:0:ffff::2%1 (Unencrypted)
	fec0:0:0:ffff::3%1 (Unencrypted)

IPv4 address:	12.168.10.4
---------------	-------------

Manufacturer:	Realtek
---------------	---------

Description:	Realtek PCIe GbE Family Controller
--------------	------------------------------------

Driver version:	10.71.312.2024
-----------------	----------------

Physical address (MAC):	20:53:8D:01:B9:C5
-------------------------	-------------------

support

Help with Ethernet

Updating network adapter or driver

network & internet > Ethernet

Unidentified network
No internet

Ethernet:

Specific DNS Suffix . . . :
IPv6 Address : fe80::98fd:b24e:c03e:8990%16
IPv4 Address : 192.168.10.3
Subnet Mask : 255.255.255.0
Default Gateway :

Ethernet 2:

Specific DNS Suffix . . . :
IPv6 Address : fe80::b5ce:7806:8555:a6c3%8
IPv4 Address : 192.168.56.1
Subnet Mask : 255.255.255.0
Default Gateway :

Local Area Connection* 1:

Media disconnected
Specific DNS Suffix . . . :

Wi-Fi:

Specific DNS Suffix . . . :
IPv6 Address : fe80::523c:8af0:568:260d%3
IPv4 Address : 192.168.10.2
Subnet Mask : 255.255.255.0
Default Gateway :

Give feedback

Search

DELL



Unidentified network
No internet

Authentication settings

Metered connection

Some apps might work differently to reduce data usage when you're connected to this network

Set a data limit to help control data usage on this network

IP assignment:	Manual
IPv4 address:	192.168.10.5
IPv4 mask:	255.255.255.0
DNS server assignment:	Automatic (DHCP)
Aggregated link speed (Receive/Transmit):	1000/1000 (Mbps)
Link-local IPv6 address:	fe80::84c9:b22d:ffba:d80b%2
IPv6 DNS servers:	fec0:0:0:ffff::1%1 (Unencrypted) fec0:0:0:ffff::2%1 (Unencrypted) fec0:0:0:ffff::3%1 (Unencrypted)
IPv4 address:	192.168.10.5
Manufacturer:	Realtek
Description:	Realtek PCIe GbE Family Controller
Driver version:	10.71.312.2024
Physical address (MAC):	20:53:8D:02:07:98

Related support



Help with Ethernet

Update network adapter or driver

Connection-specific DNS Suffix . : ctutaining.co.za
Link-local IPv6 Address : fe80::8989:71b4:9cc9:3aee%12
IPv4 Address. : 172.16.3.170

Find a setting

Network & internet > Wi-Fi > CTU-Staff

☒ Connect automatically when in range

Network profile type

☒ **Public network (Recommended)**

Your device is not discoverable on the network. Use this in most cases—when connected to a network at home, work, or in a public place.

☐ **Private network**

Your device is discoverable on the network. Select this if you need file sharing or use apps that communicate over this network. You should know and trust the people and devices on the network.

Configure firewall and security settings

Metered connection

Some apps might work differently to reduce data usage when you're connected to this network

Set a data limit to help control data usage on this network

Random hardware addresses

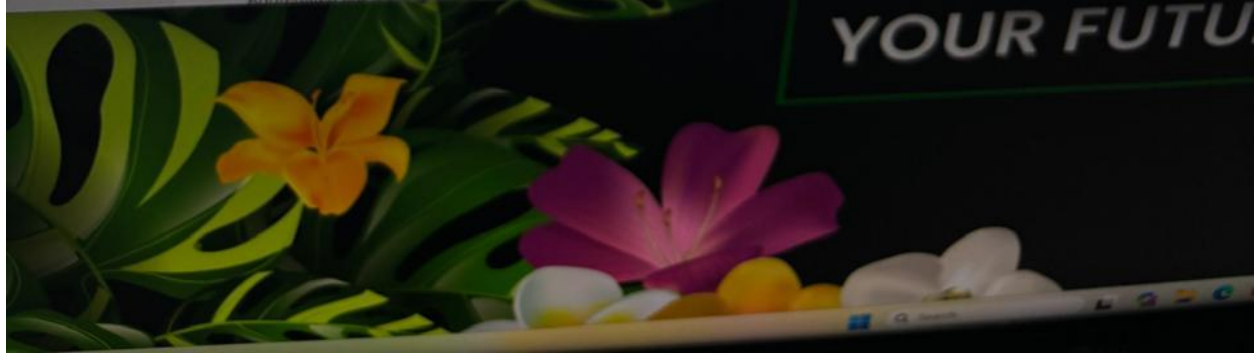
Help protect your privacy by making it harder for people to track your device location when you connect to this network. The setting takes effect the next time you connect to this network.

IP assignment:	Manual
IPv4 address:	192.168.20.2
IPv4 mask:	255.255.255.0
DNS server assignment:	Automatic (DHCP)
SSID:	CTU-Staff
Protocol:	Wi-Fi 6 (802.11ax)
Security type:	WPA2-Personal
Manufacturer:	Intel Corporation
Description:	Intel(R) Wi-Fi 6 AX201 160MHz
Driver version:	24.20.0.4
Network band (channel):	5 GHz (128)
Aggregated link speed (802.11ac):	2.502/2.502 Gbps

Off

Conn
Conne
Link-l
IPv4 A
Subnet
Default
Wireless LAN
Media Stat
Connection
Ethernet adapt
Connection-s
Link-local IP
Autoconfigura
Subnet Mask
Default Gateway
Wireless LAN adapt
Connection-speci
Link-local IPv6 A
IPv4 Address
Subnet Mask
Default Gateway
Ethernet adapter vEther
Connection-specific

YOUR FUTU



Find a setting

Network & internet > Wi-Fi > CTU-Staff

Network profile type

☒ **Public network (Recommended)**

Your device is not discoverable on the network. Use this in most cases—when connected to a network at home, work, or in a public place.

☐ **Private network**

Your device is discoverable on the network. Select this if you need file sharing or use apps that communicate over this network with people and devices on the network.

Configure firewall and security settings

Metered connection

Some apps might work differently to reduce data usage when you're connected to this network.

Set a data limit to help control data usage on this network

Random hardware addresses

Help protect your privacy by making it harder for people to track your device location when you connect to this network. The setting takes effect the next time you connect to this network.

IP assignment:	Manual
IPv4 address:	192.168.20.3
IPv4 mask:	255.255.255.0
DNS server assignment:	Automatic (DHCP)
SSID:	CTU-Staff
Protocol:	Wi-Fi 6 (802.11ax)
Security type:	WPA2-Personal
Manufacturer:	Intel Corporation
Description:	Intel(R) Wi-Fi 6 AX201 160MHz
Driver version:	24.50.0.4
Network band (channel):	5 GHz (128)
Aggregated link speed (Receive/Transmit):	2402/2402 (Mbps)
Link-local IPv6 address:	fe80:354b:db7e:c8f4:f925%15
IPv6 DNS servers:	fec0:0:0:ffff:1%1 (Unencrypted) fec0:0:0:ffff:2%1 (Unencrypted) fec0:0:0:ffff:3%1 (Unencrypted)
IPv4 address:	192.168.20.3
Physical address (MAC):	64:4A:7D:5B:D7:04

Advanced Wi-Fi network properties

Related support

 [Help from the web](#)

Updating wireless network driver

Find a setting

Network & internet > Ethernet



Ethernet
Not connected

Authentication settings

Metered connection

Some apps might work differently to reduce data usage when you're connected to this network

Set a data limit to help control data usage on this network

IP assignment:	Manual
IPv4 address:	192.168.10.2
IPv4 mask:	255.255.255.0

DNS server assignment:	Automatic (DHCP)
------------------------	------------------

Manufacturer:	Realtek
Description:	Realtek PCIe GbE Family Controller
Driver version:	10.71.312.2024
Physical address (MAC):	20:53:8D:39:BC:31



Give feedback

Vlan	IP-Address
Vlan10	unassigned
Vlan20	192.168.10.1
Vlan30	192.168.20.1
GigabitEthernet0/1	192.168.30.1
GigabitEthernet0/2	unassigned
GigabitEthernet0/3	unassigned
GigabitEthernet0/4	unassigned
GigabitEthernet0/5	unassigned
GigabitEthernet0/6	unassigned
GigabitEthernet0/7	unassigned
GigabitEthernet0/8	unassigned
GigabitEthernet0/9	unassigned
GigabitEthernet0/10	unassigned
GigabitEthernet0/11	unassigned
GigabitEthernet0/12	unassigned
GigabitEthernet0/13	unassigned
GigabitEthernet0/14	unassigned
GigabitEthernet0/15	unassigned
GigabitEthernet0/16	unassigned
GigabitEthernet0/17	unassigned
GigabitEthernet0/18	unassigned

YES unset up
YES manual up
YES manual up
YES manual up
YES unset down
YES unset up
YES unset up
YES unset up
YES unset up
YES unset up
YES unset up
YES unset down
YES unset up
YES unset down
S unset down
S unset down
? unset down
unset down
unset down
unset down

up
up
up
down
up
up
up
up
up
down
up
down
down
down
down
down
down



~~mdm@ctucareer~~

mdm@ctucareer.co.za

CTU@2026

PATCH PANEL SET UP

4	PC 3	PC 5	
3	PC 2	PC 1	
2	PC 9	PC 6	PC 4
1	PC 7	PC 10	PC 8

1) 10.2

2) 10.3

3) 10.6

4) 10.5

5) 20.2

6) 20.3

7) 20.6

8) 20.5

9) 30.2

10) 30.3

Vlan 10 (engineering/management) PC 1-4
Vlan 20 (Sales/Operations) PC 5-8
Vlan 30 (Server/Admin) PC 9-10

<u>PC 1-4: 192.168.10.0</u>	PC 1	PC 2	PC 3	PC 4
<u>PC 5-8: 192.168.20.0</u>		PC 7	PC 6	PC 5
<u>PC 9-10: 192.168.30.0</u>	BOX			

	PC 8	PC 10
	PC 7	PC 9

identified network
internet

Authentication settings

Metered connection

Some apps might work differently to reduce data usage when you're connected to this network

[Set a data limit to help control data usage on this network](#)

IP assignment:	Manual
IPv4 address:	192.168.20.4
IPv4 mask:	255.255.255.0

DNS server assignment:	Automatic (DHCP)
------------------------	------------------

Aggregated link speed (Receive/ Transmit):	1000/1000 (Mbps)
---	------------------

Link-local IPv6 address:	fe80::f37a:ec6b:2f97:8b8a%19
--------------------------	------------------------------

IPv6 DNS servers:	fec0:0:0:ffff::1%1 (Unencrypted) fec0:0:0:ffff::2%1 (Unencrypted) fec0:0:0:ffff::3%1 (Unencrypted)
-------------------	--

IPv4 address:	192.168.20.4
---------------	--------------

Manufacturer:	Realtek
---------------	---------

Description:	Realtek PCIe GbE Family Controller
--------------	------------------------------------

Driver version:	10.71.312.2024
-----------------	----------------

Physical address (MAC):	20:53:8D:02:07:85
-------------------------	-------------------

 [Give feedback](#)

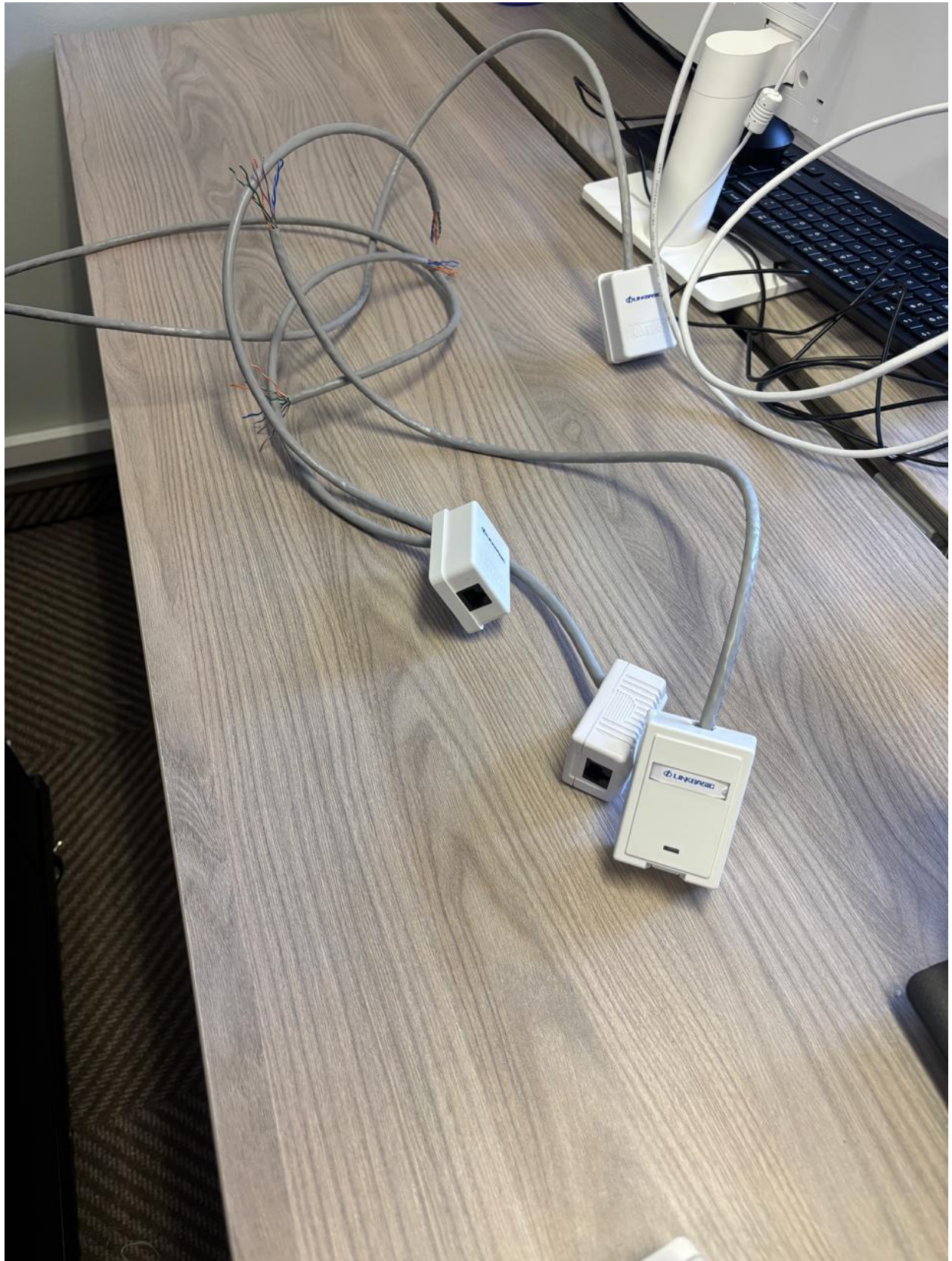
COM3 - PuTTY

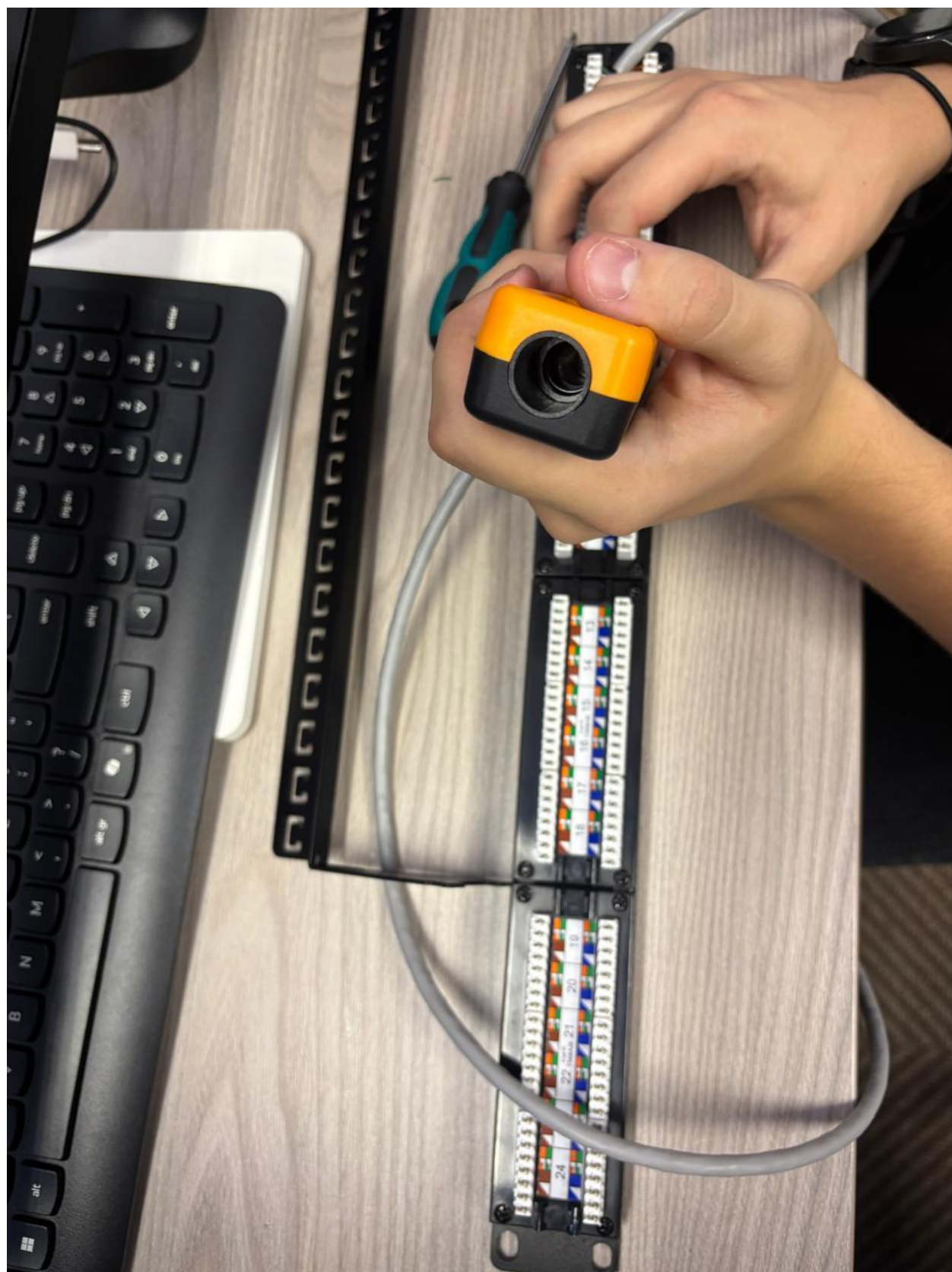
```
00:16:37: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/10, c
hanged state to down
00:16:38: %LINK-3-UPDOWN: Interface GigabitEthernet0/10, changed state to down
00:16:42: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/8, ch
anged state to down
00:16:43: %LINK-3-UPDOWN: Interface GigabitEthernet0/8, changed state to down
Switch(config-vlan)#
Switch(config-vlan)#
Switch(config-vlan)#
Switch(config-vlan)#name
00:16:57: %LINK-3-UPDOWN: Interface GigabitEthernet0/10, changed state to up
00:16:58: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/10, changed state to
% Incomplete command.

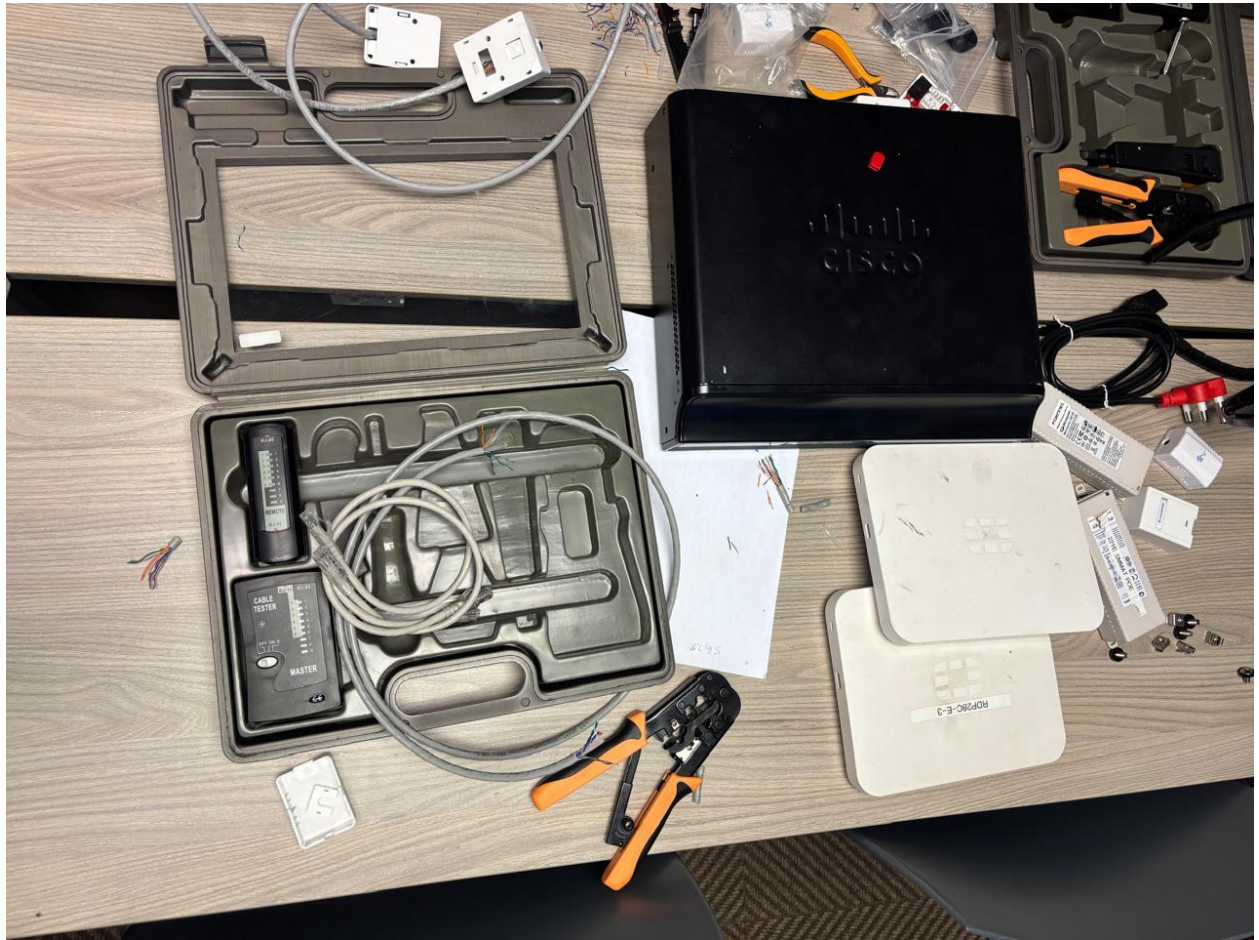
Switch(config-vlan)#
Switch(config-vlan)#exit
Switch(config)#v1
00:17:14: %LINK-3-UPDOWN: Interface GigabitEthernet0/8, changed state to up
00:17:15: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/8, changed state to upan
% Incomplete command.

Switch(config)#vlan 10
Switch(config-vlan)#name engineering/management
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name sales/operations
Switch(config-vlan)#exit
Switch(config)#vlan 30
Switch(config-vlan)#name server/admin
Switch(config-vlan)#exit
Switch(config)#
Switch(config)#exit
Switch#show 1
00:18:13: %SYS-5-CONFIG I: Configured from console by consolep in
Vlan1 is up, line protocol is up
Internet protocol processing disabled
GigabitEthernet0/1 is down, line protocol is down
Inbound access list is not set
GigabitEthernet0/2 is up, line protocol is up
Inbound access list is not set
GigabitEthernet0/3 is up, line protocol is up
Inbound access list is not set
GigabitEthernet0/4 is up, line protocol is up
Inbound access list is not set
GigabitEthernet0/5 is up, line protocol is up
Inbound access list is not set
GigabitEthernet0/6 is up, line protocol is up
Inbound access list is not set
GigabitEthernet0/7 is up, line protocol is up
Inbound access list is not set
GigabitEthernet0/8 is up, line protocol is up
Inbound access list is not set
GigabitEthernet0/9 is down, line protocol is down
Inbound access list is not set
GigabitEthernet0/10 is up, line protocol is up
Inbound access list is not set
GigabitEthernet0/11 is down, line protocol is down
Inbound access list is not set
GigabitEthernet0/12 is down, line protocol is down
Inbound access list is not set
GigabitEthernet0/13 is down, line protocol is down
```







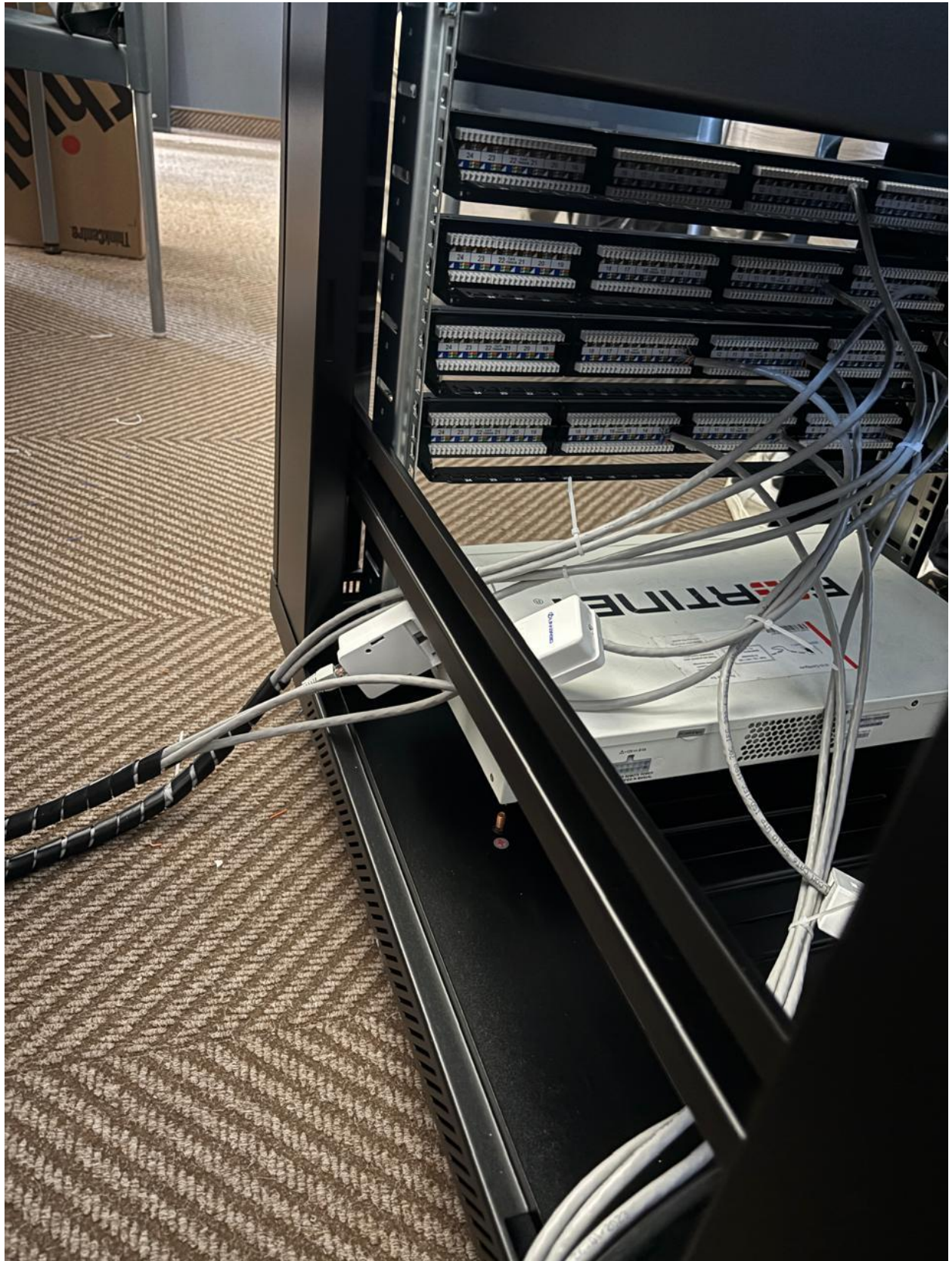


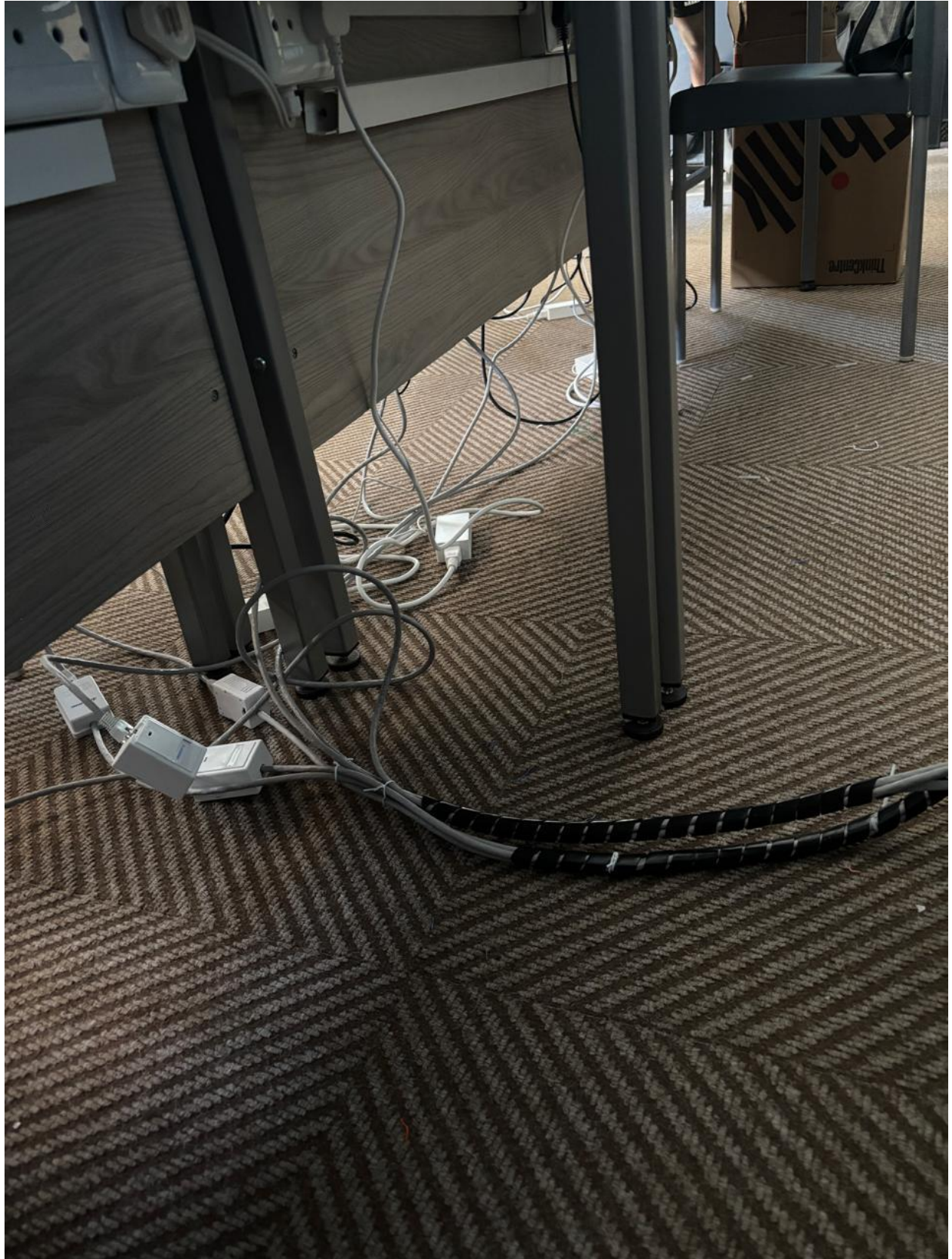






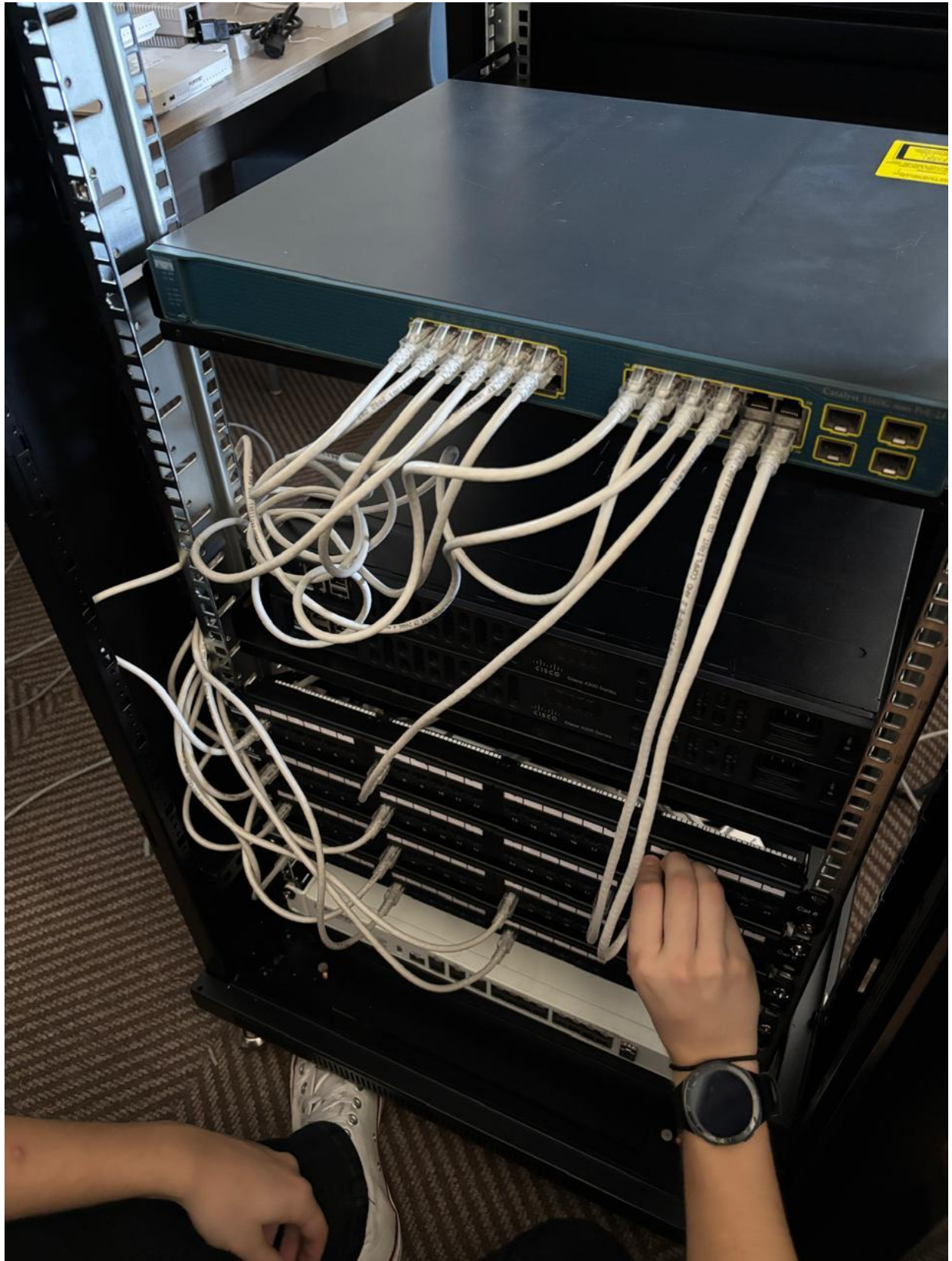


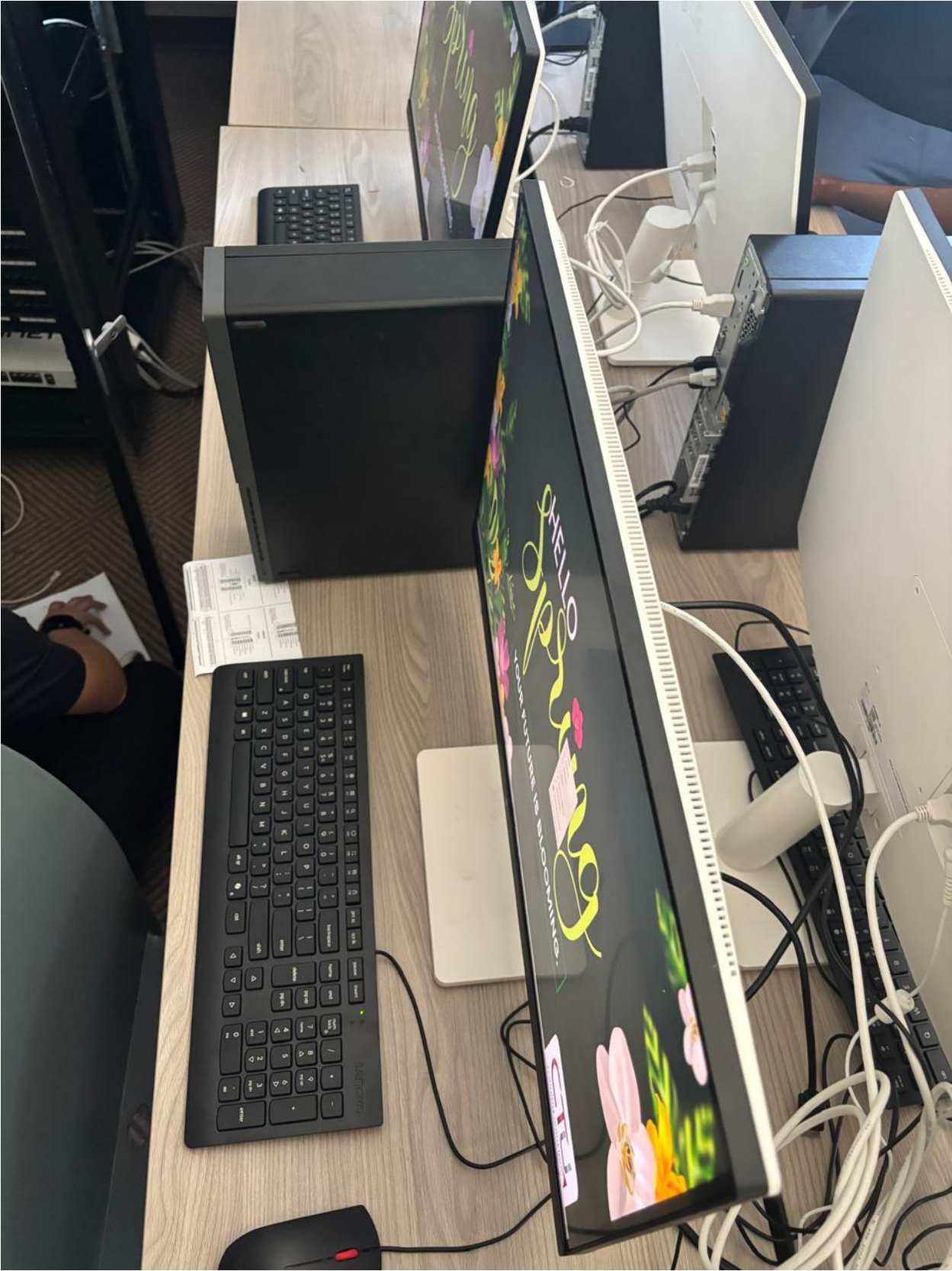


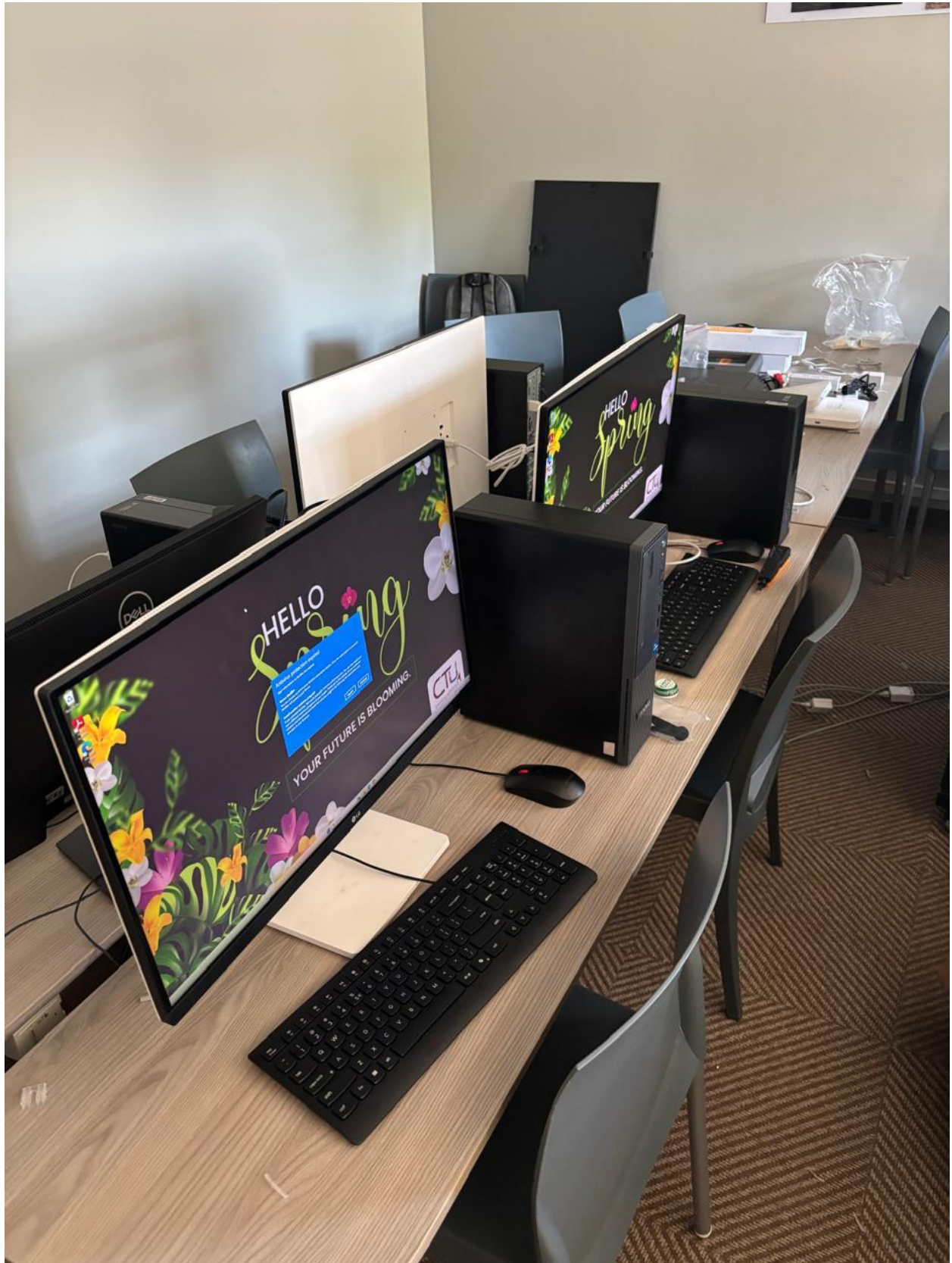






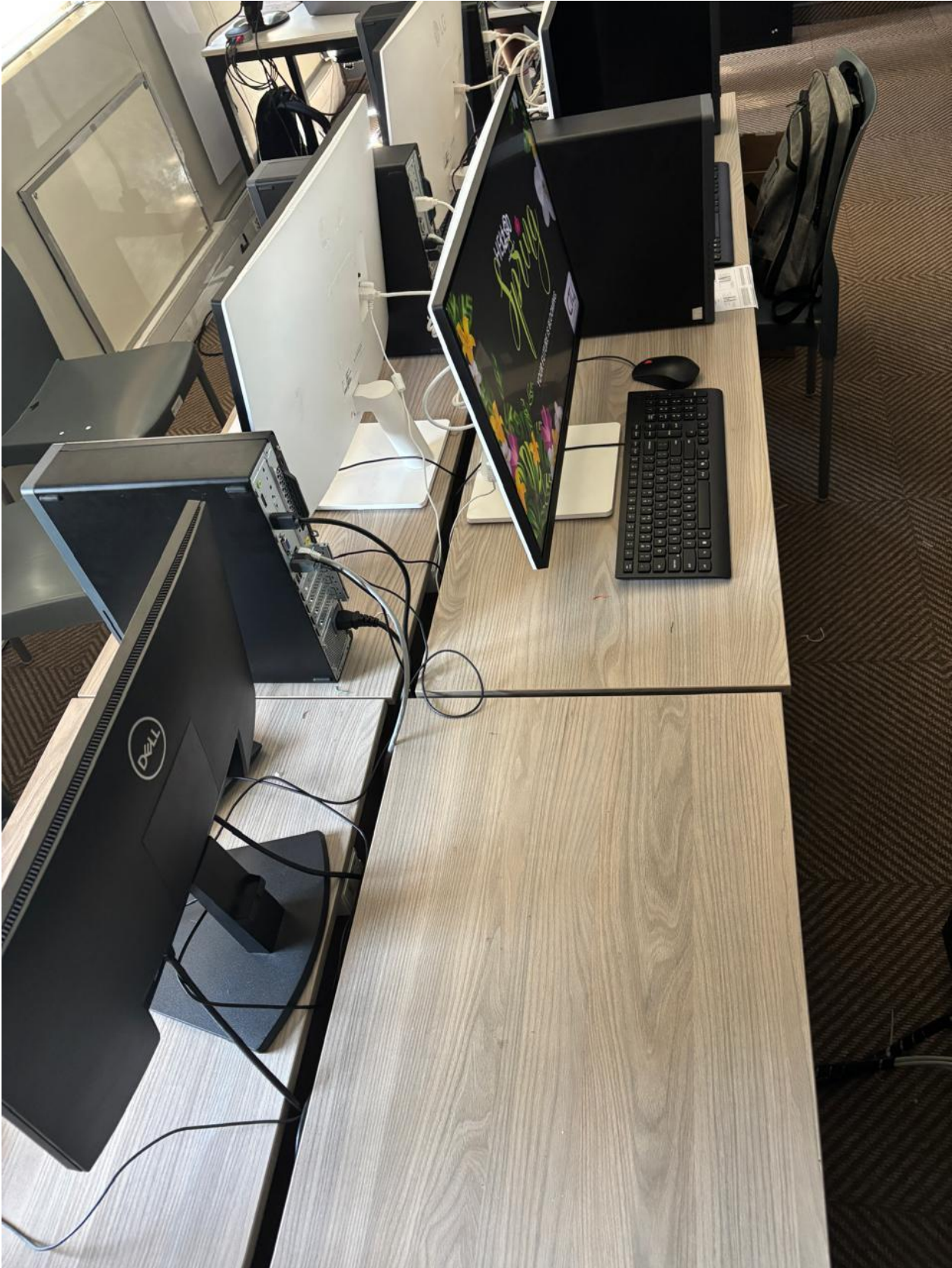




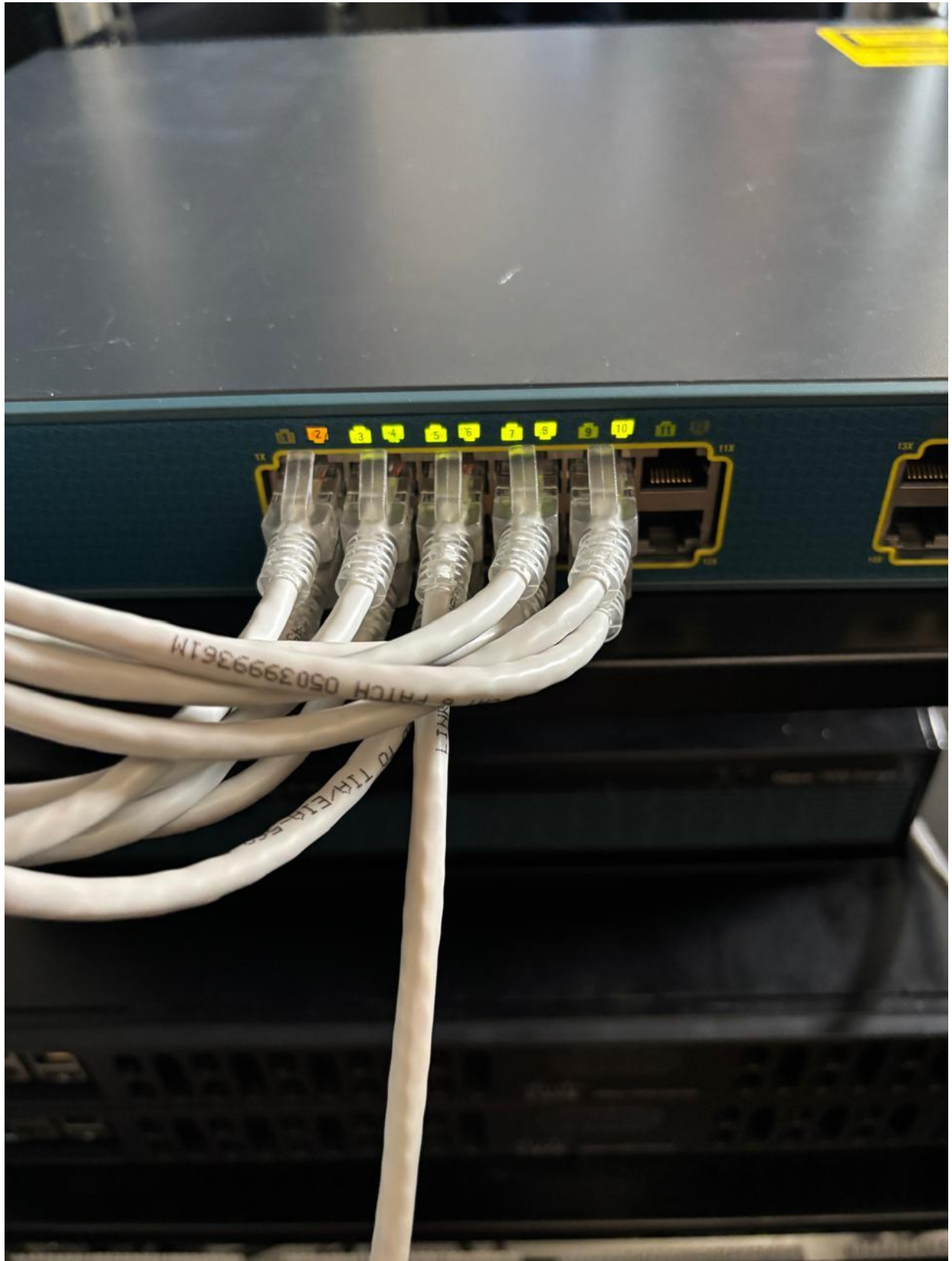
















10.2a

9) 30.2

10) 30.3

1) 10.2

2) 10.3

3) 10.4

4) 10.5

~~5) 20.2~~

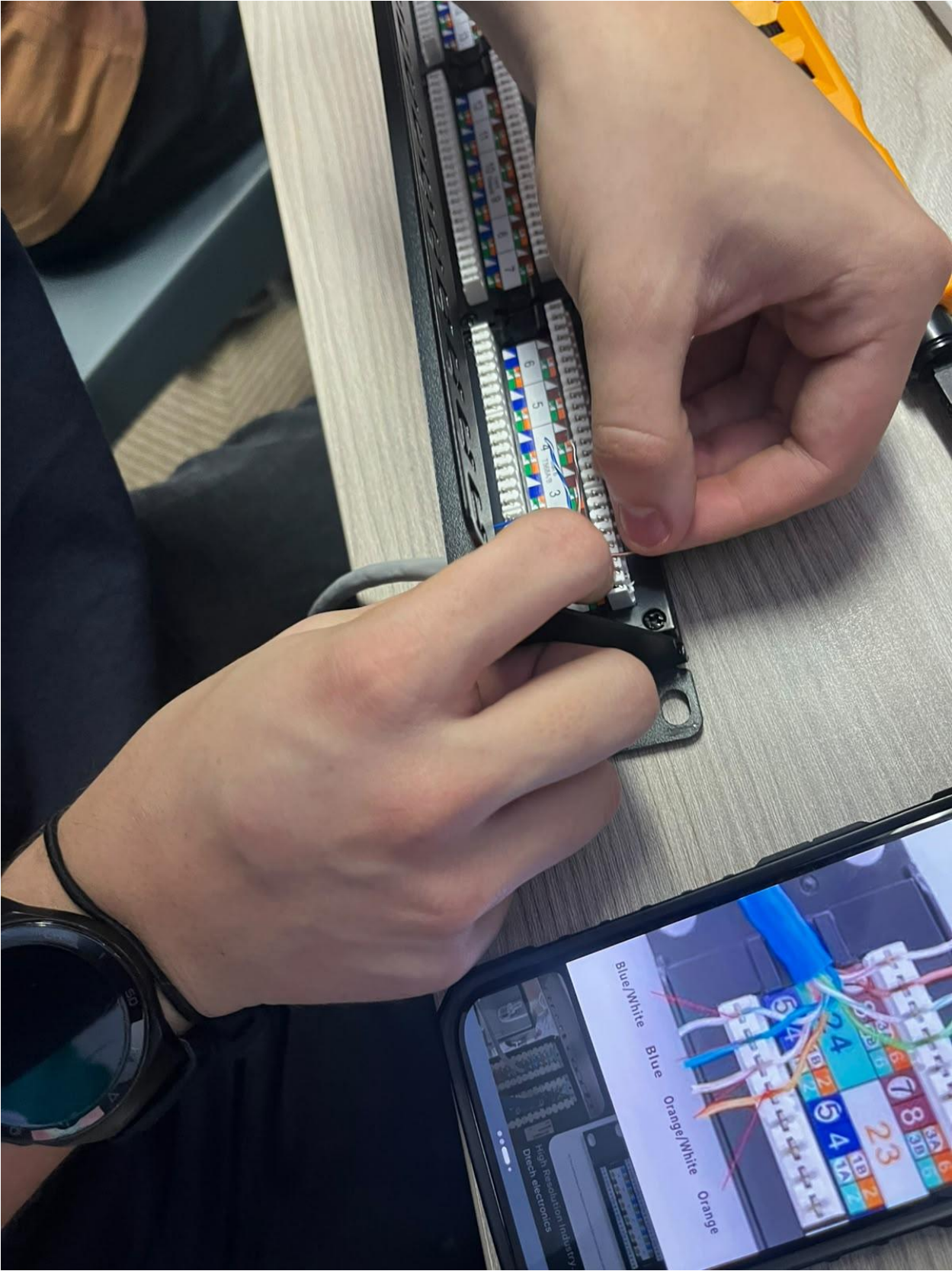
6) 20.3

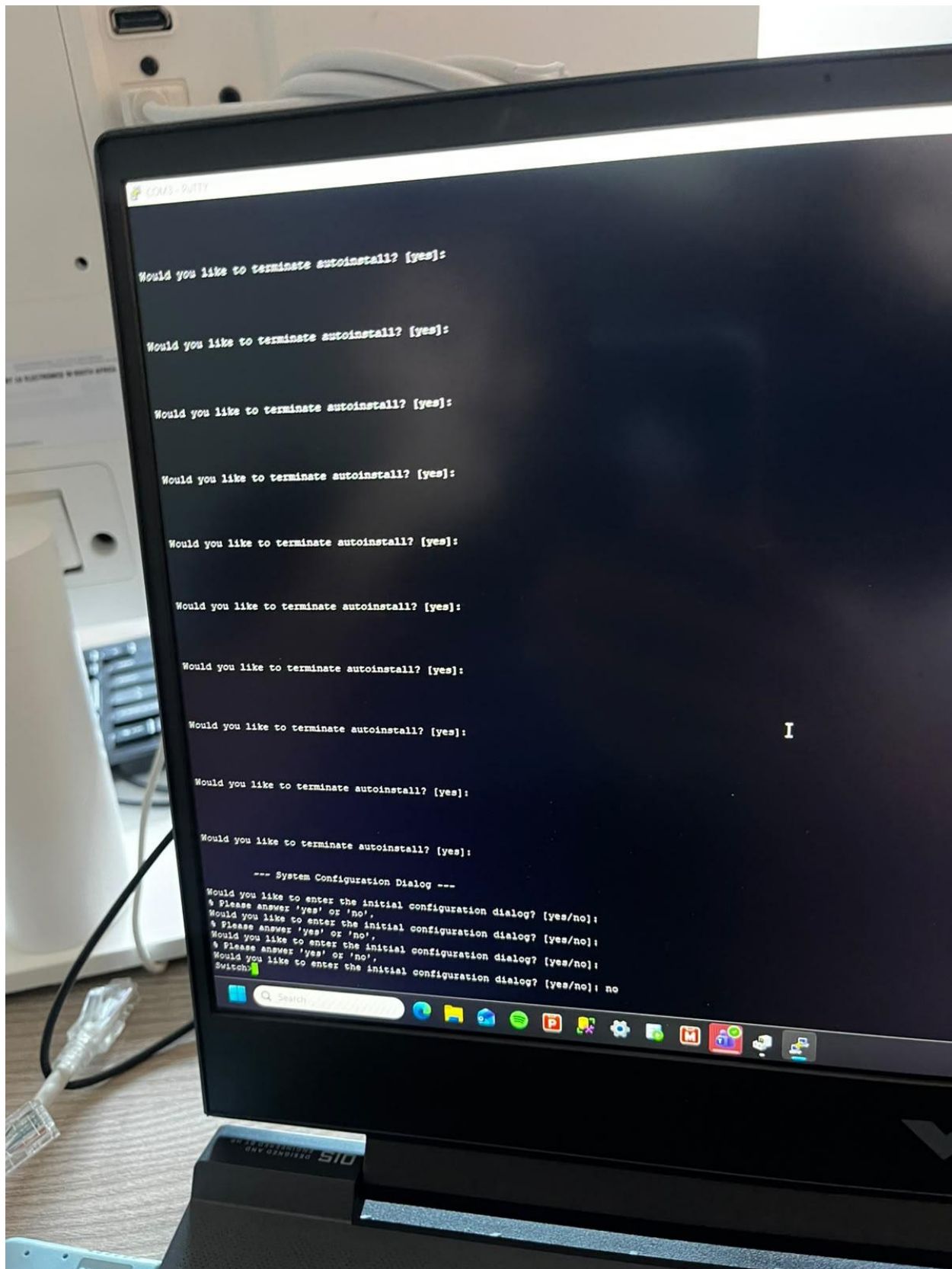
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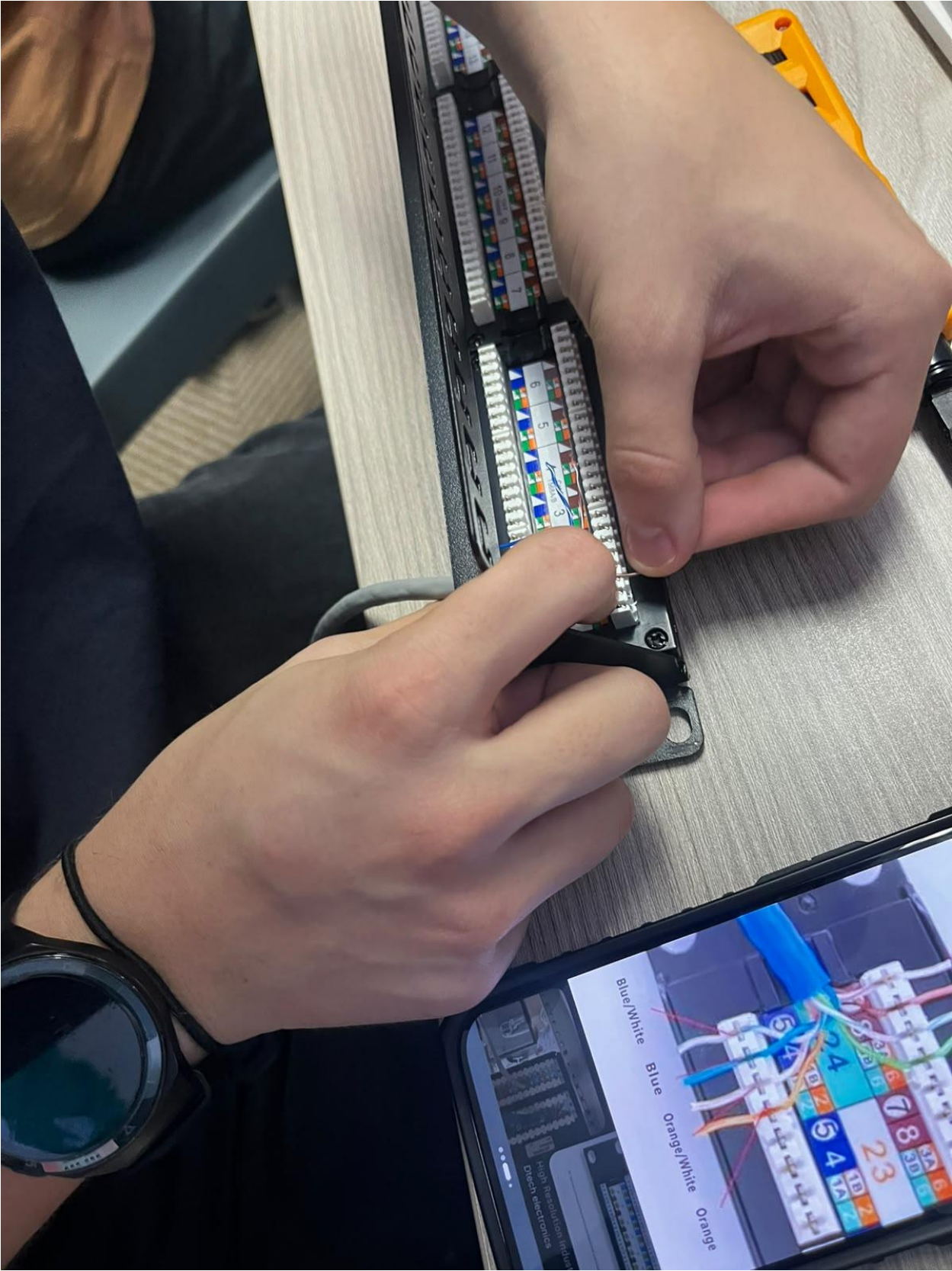
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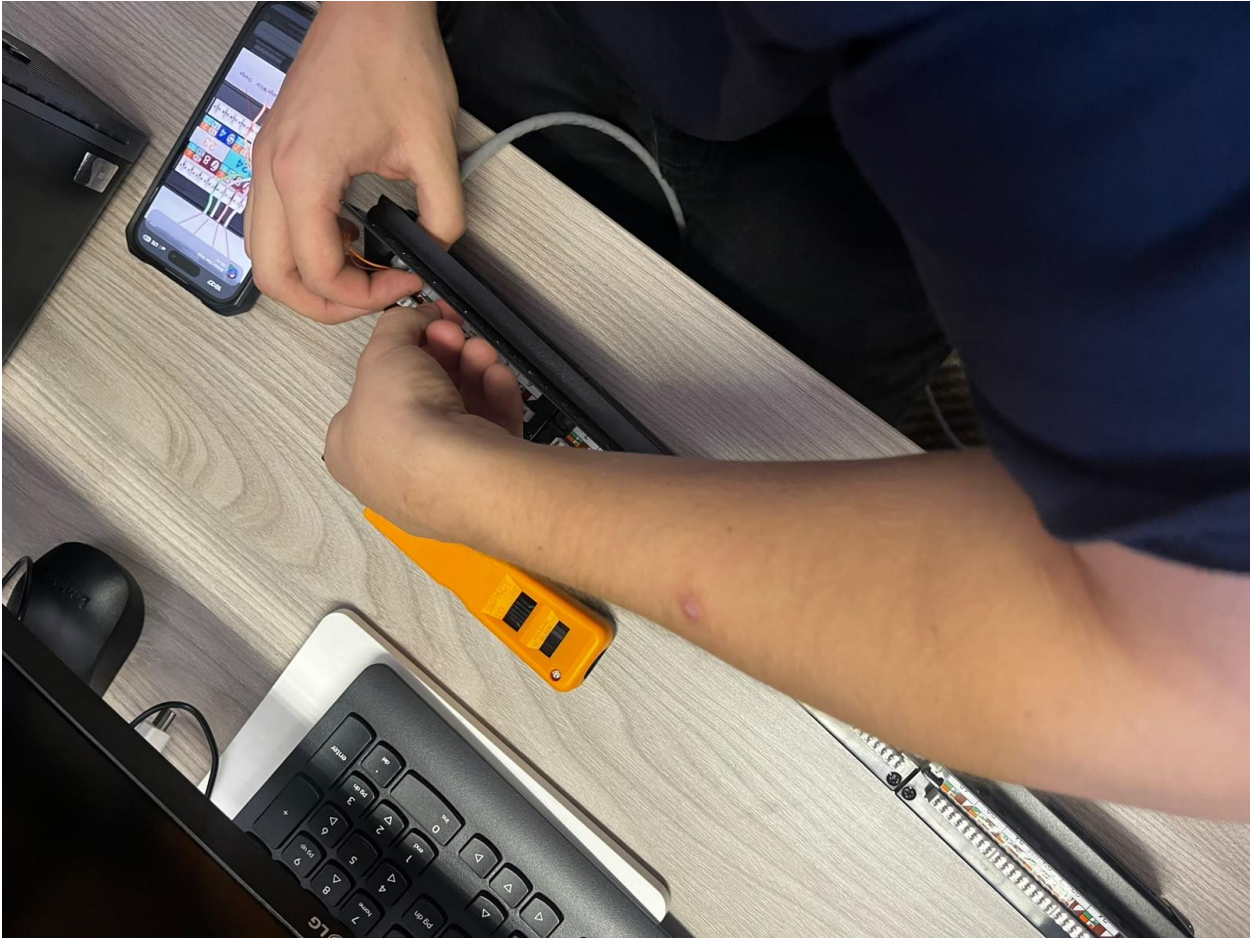




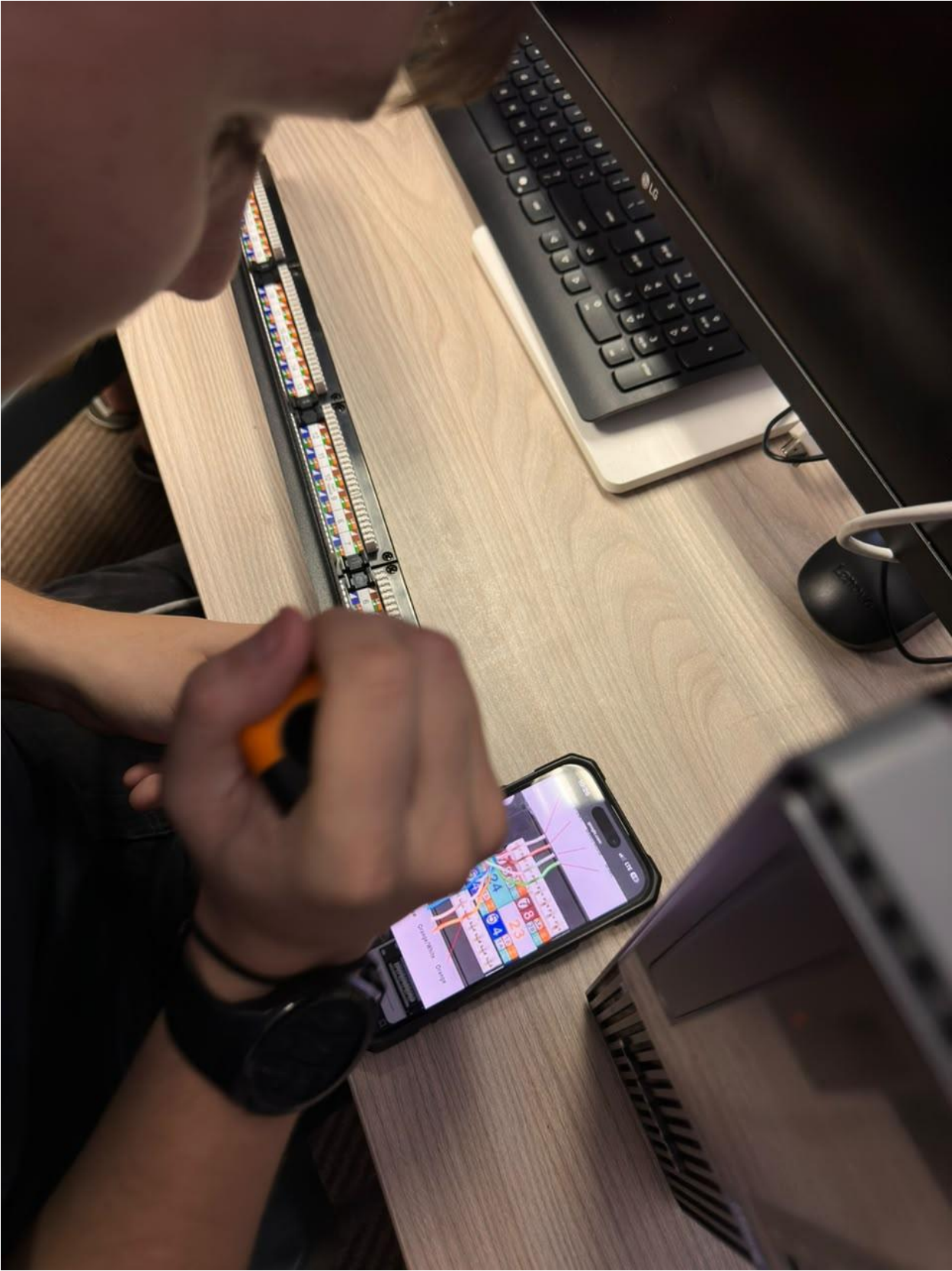




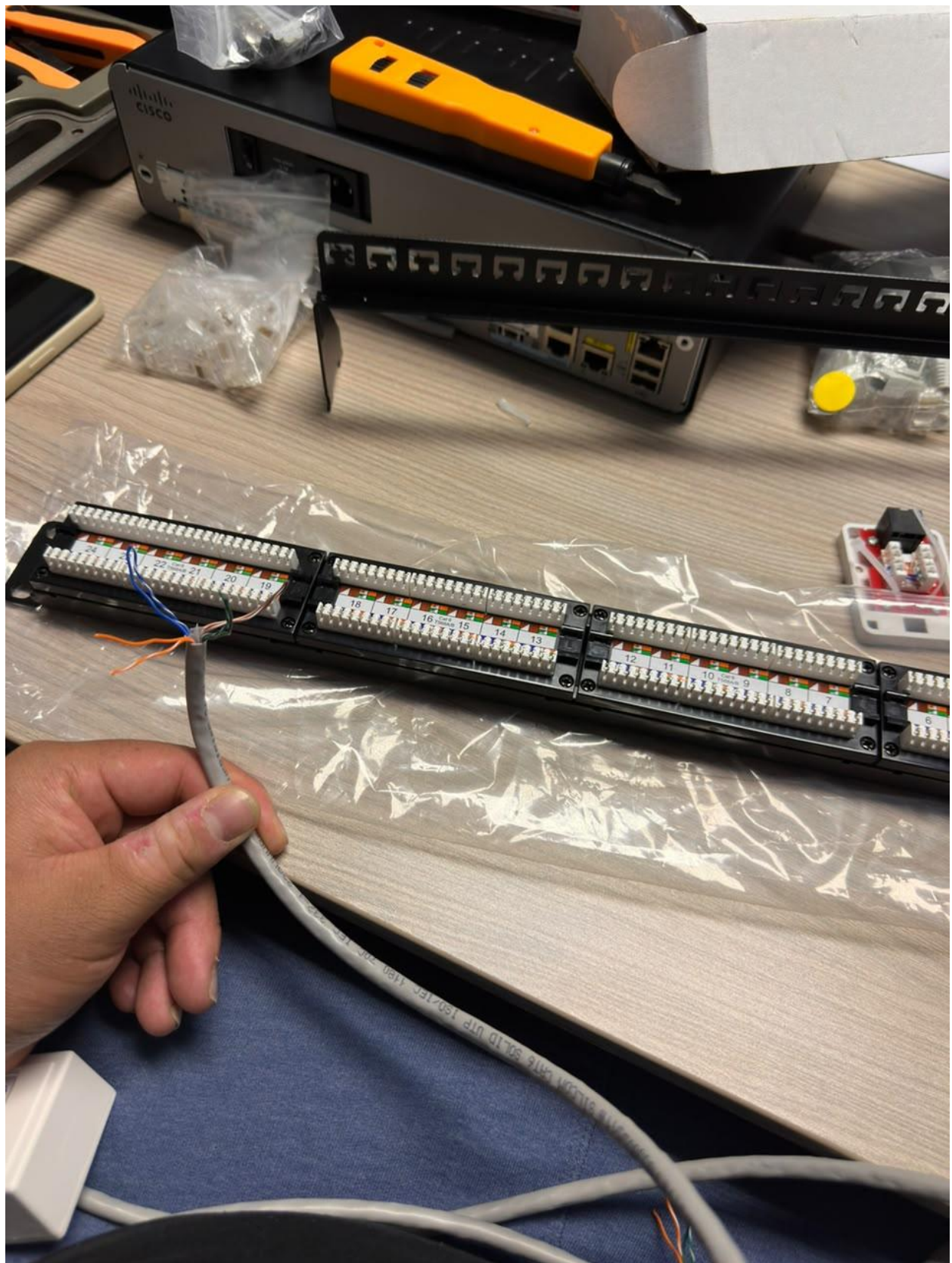


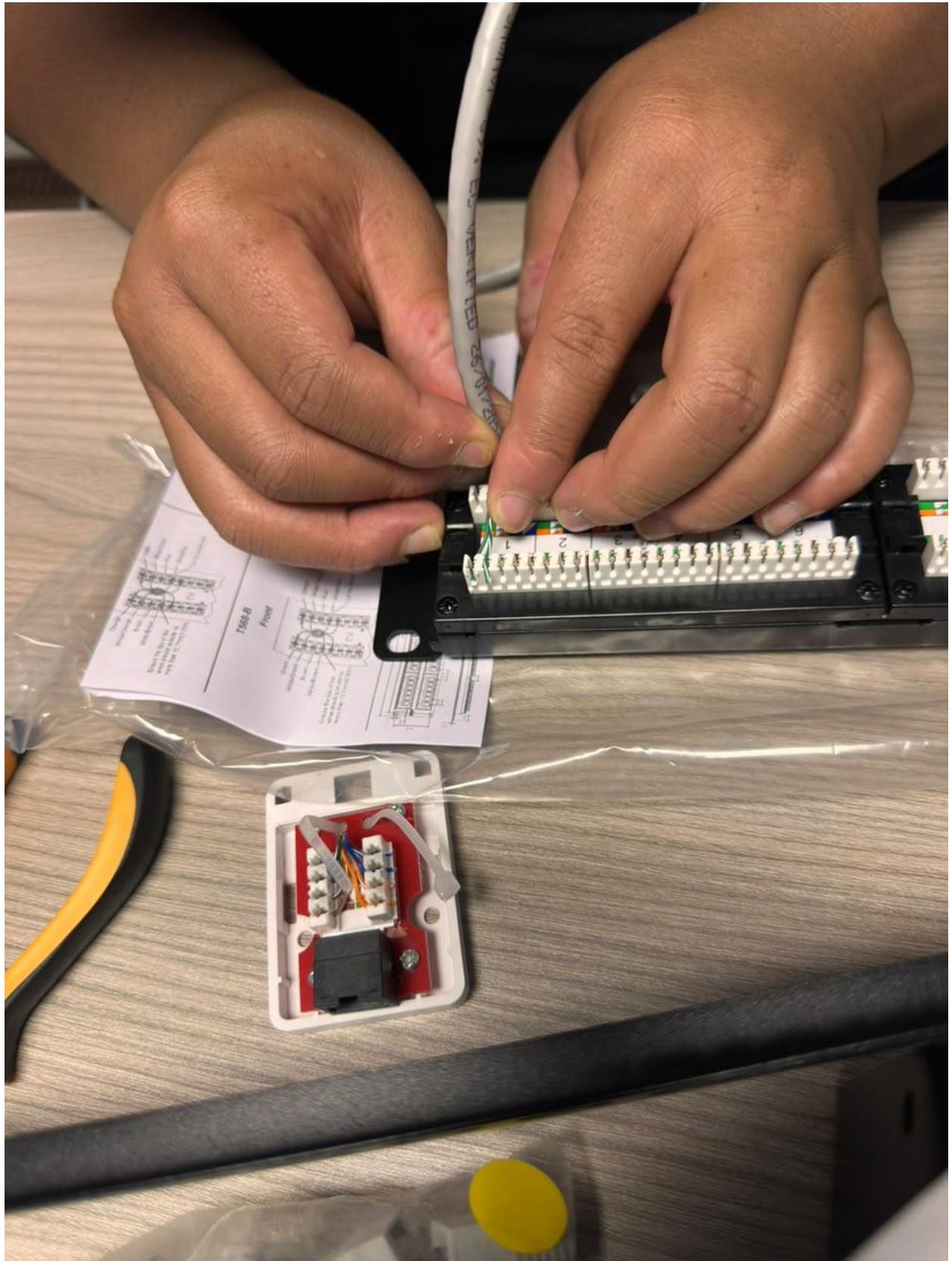


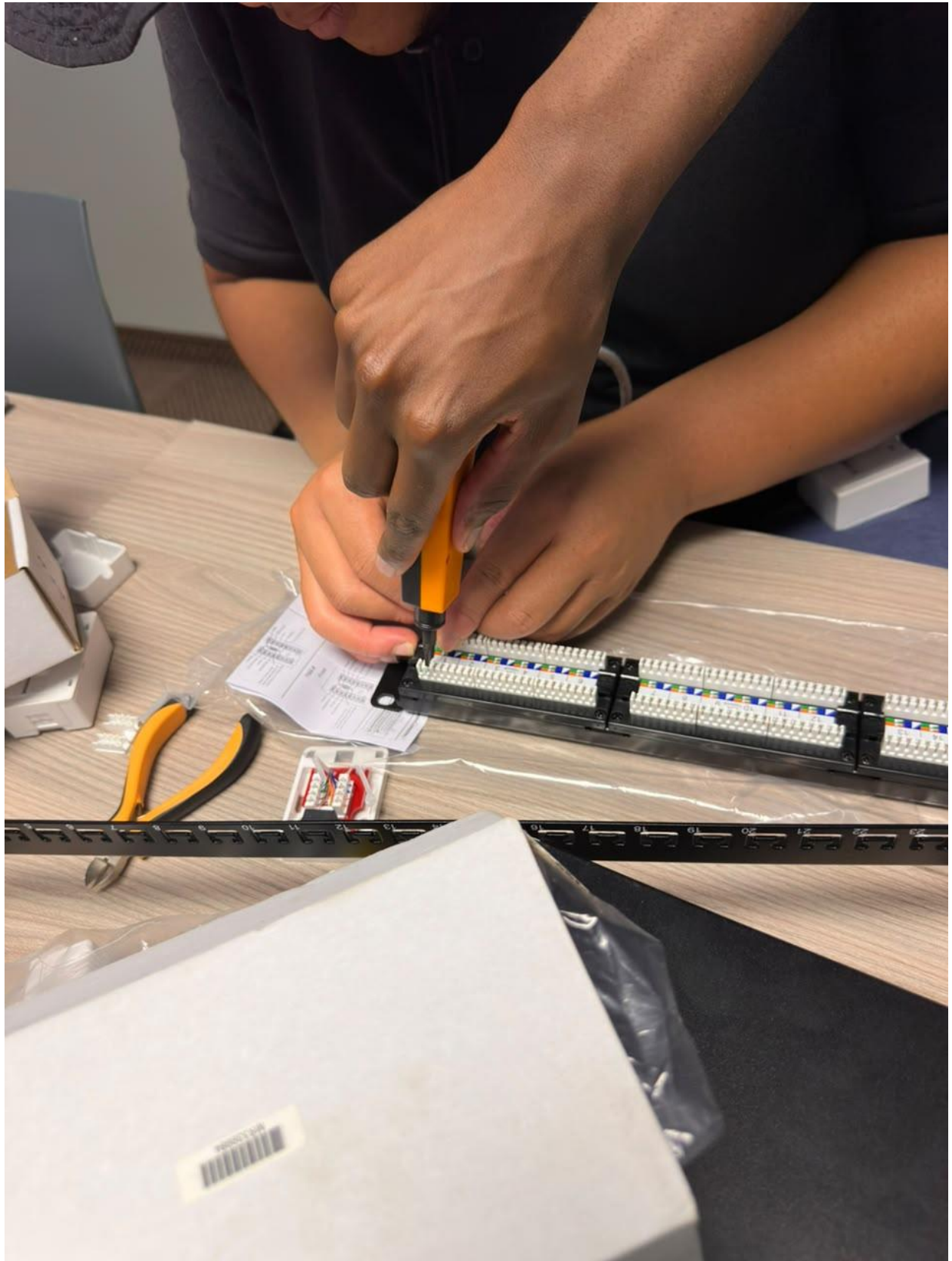


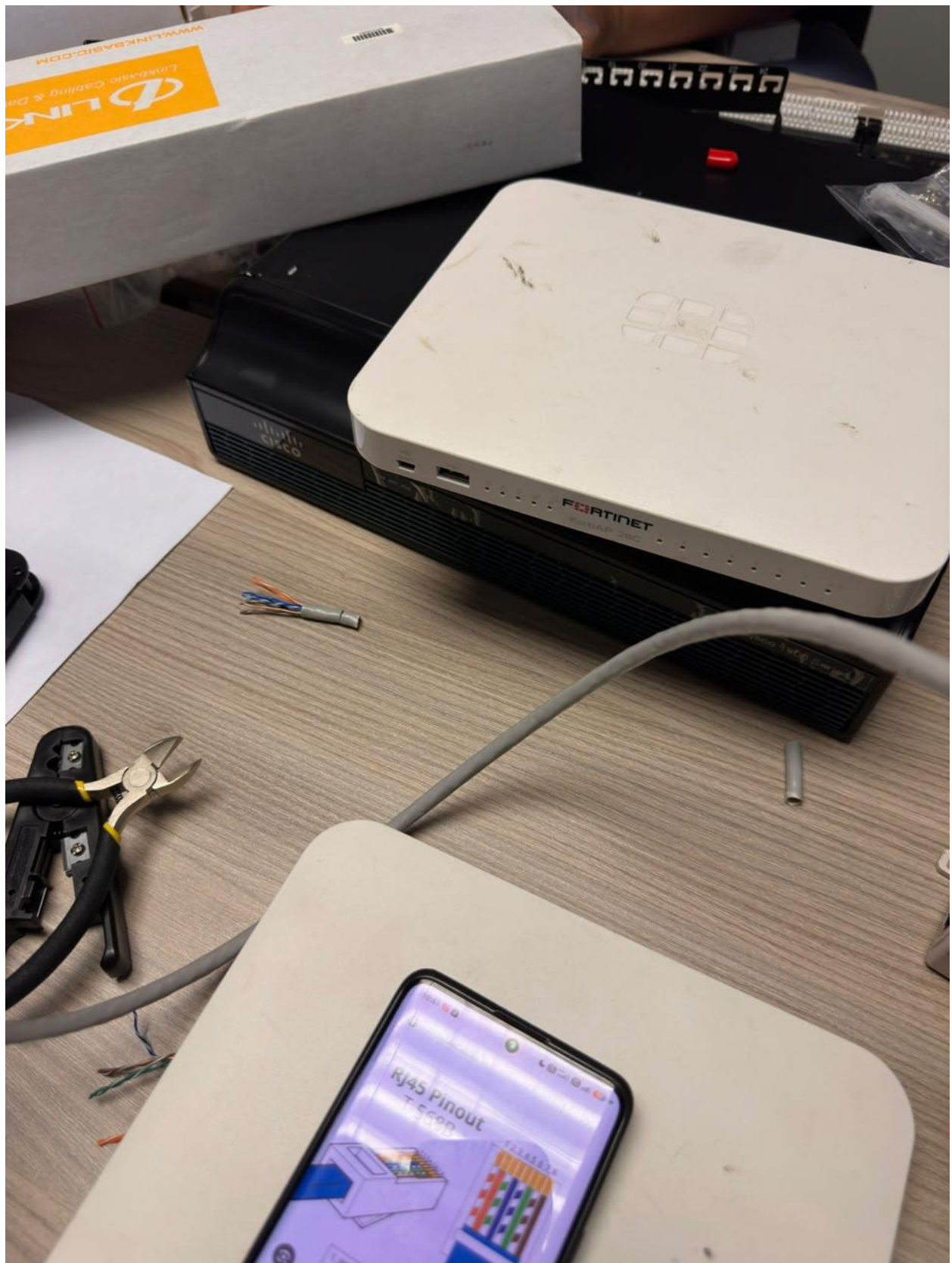




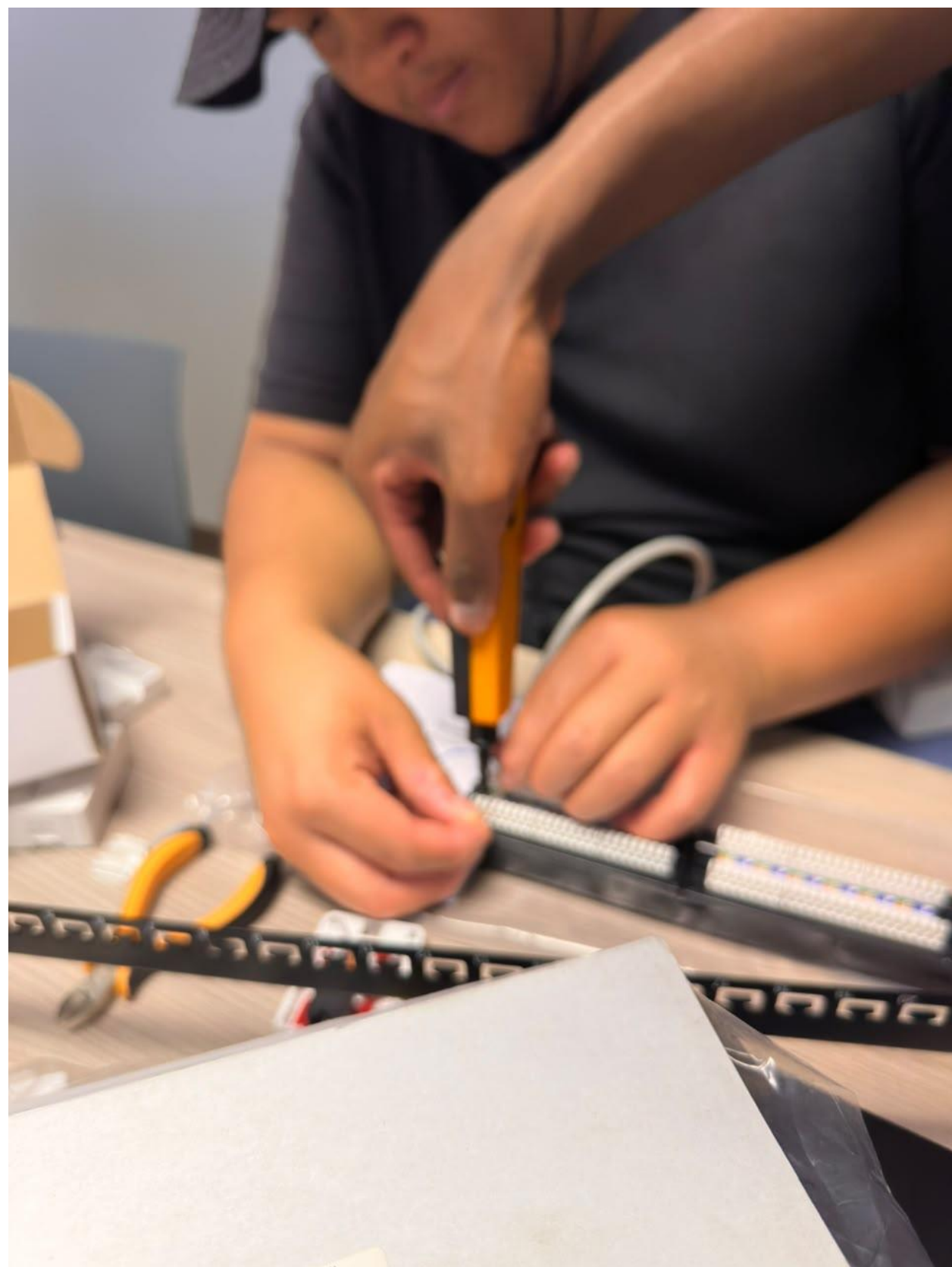


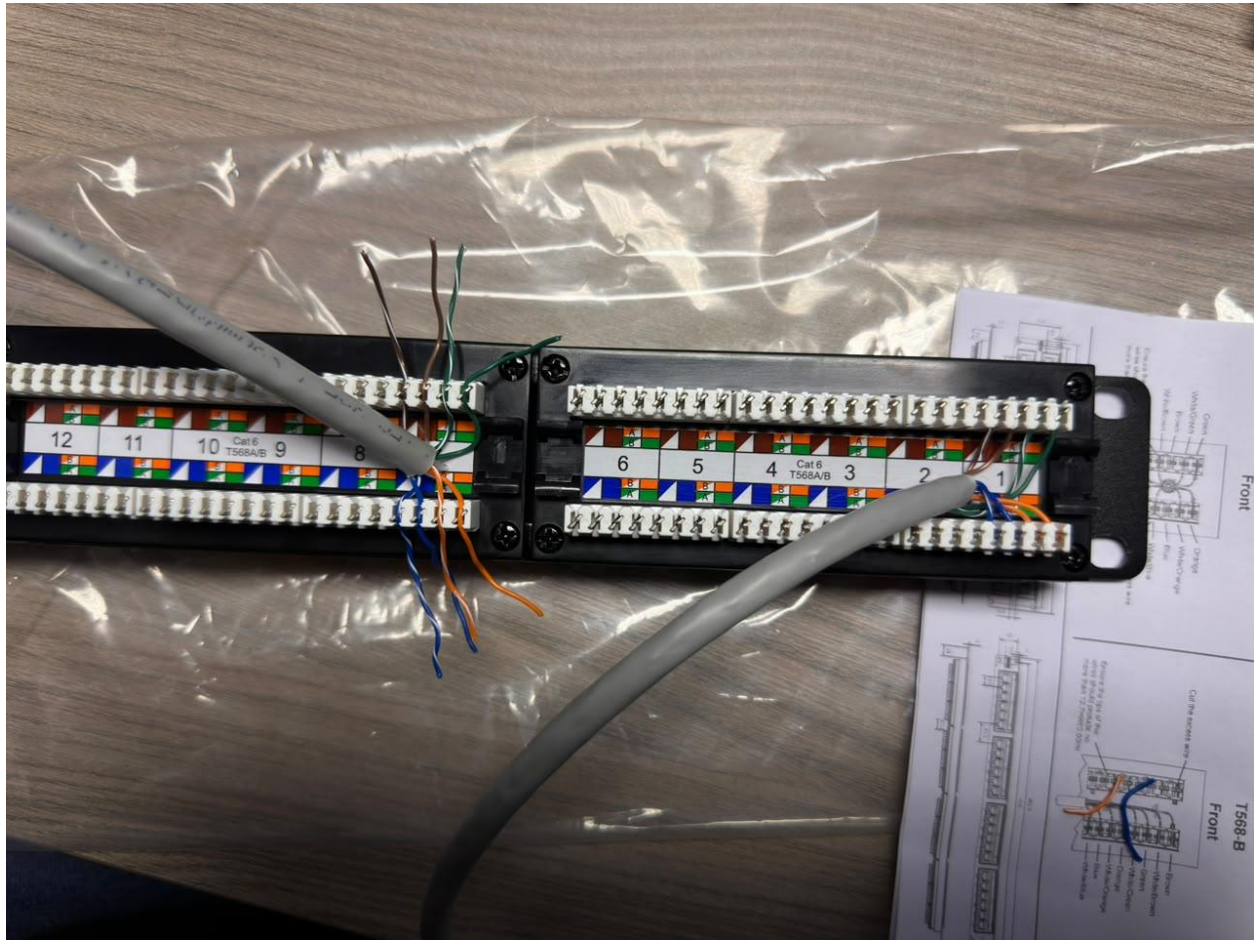


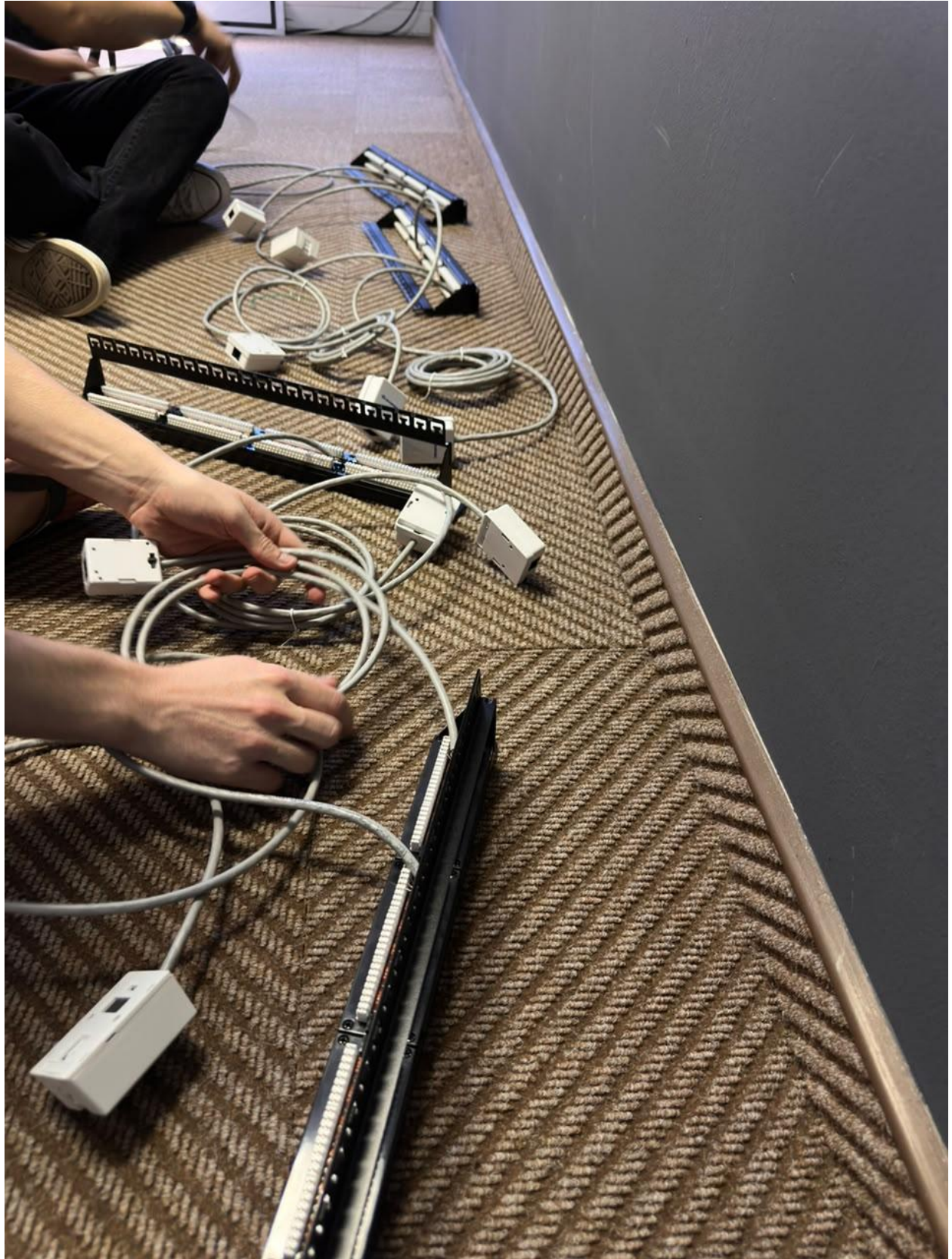












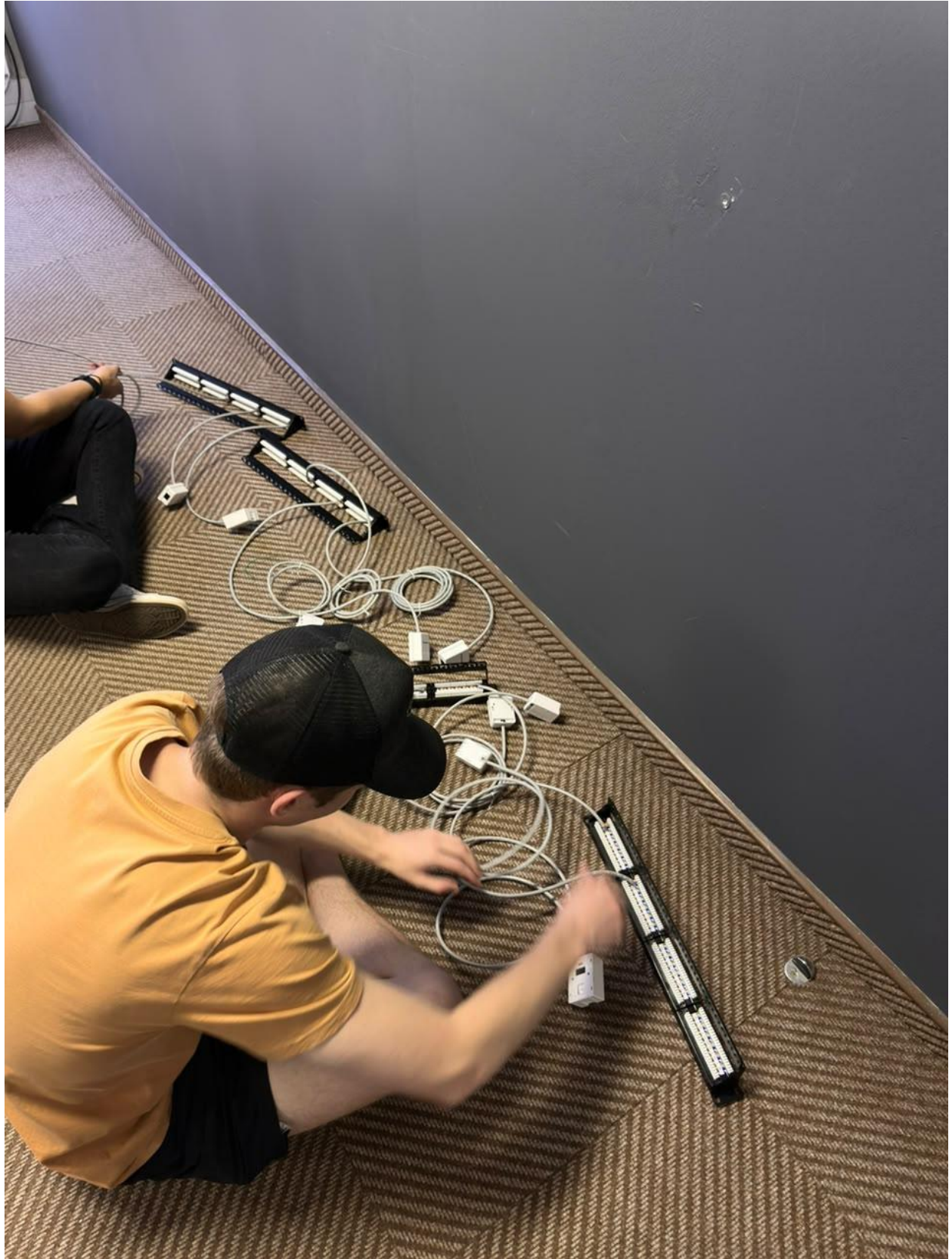


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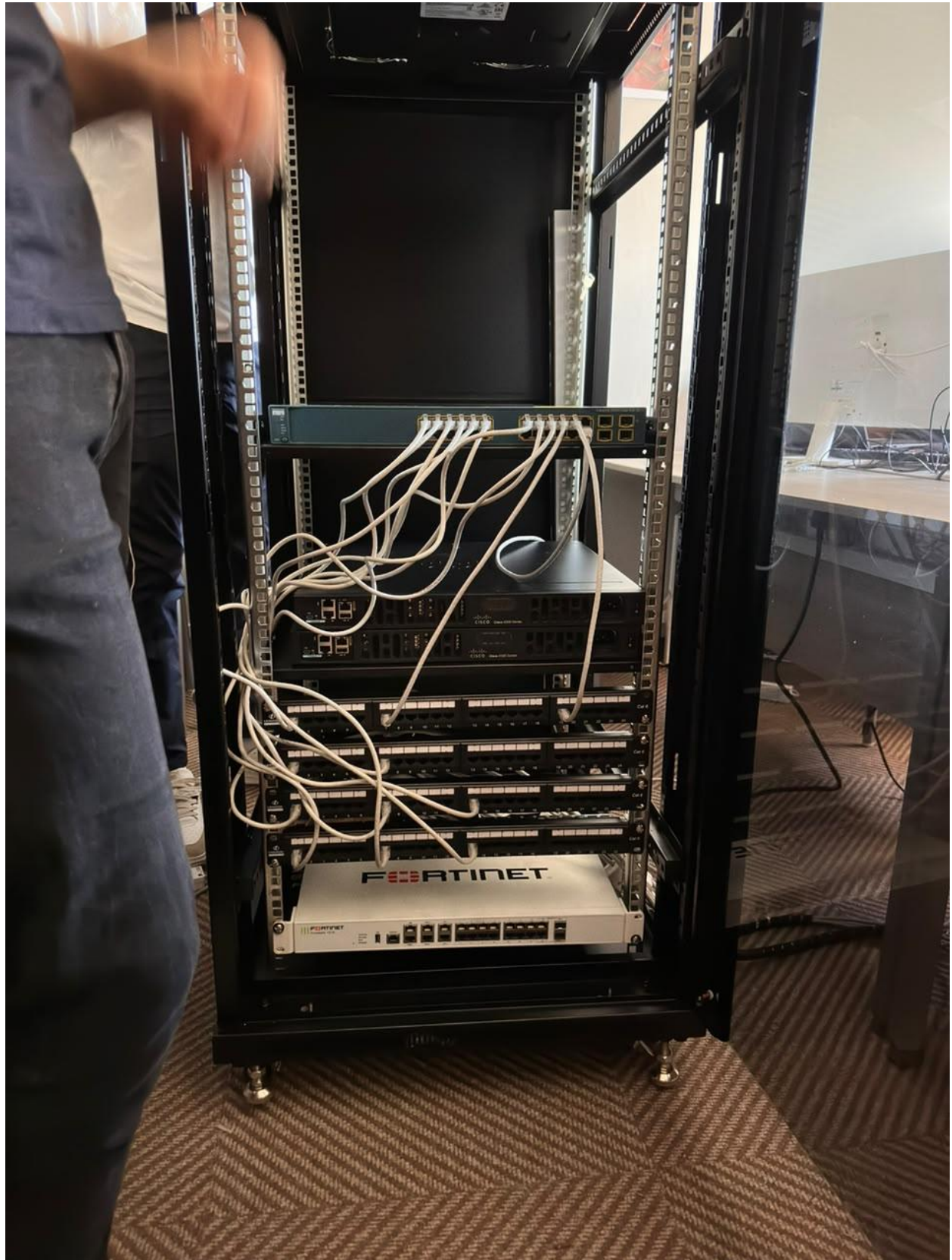
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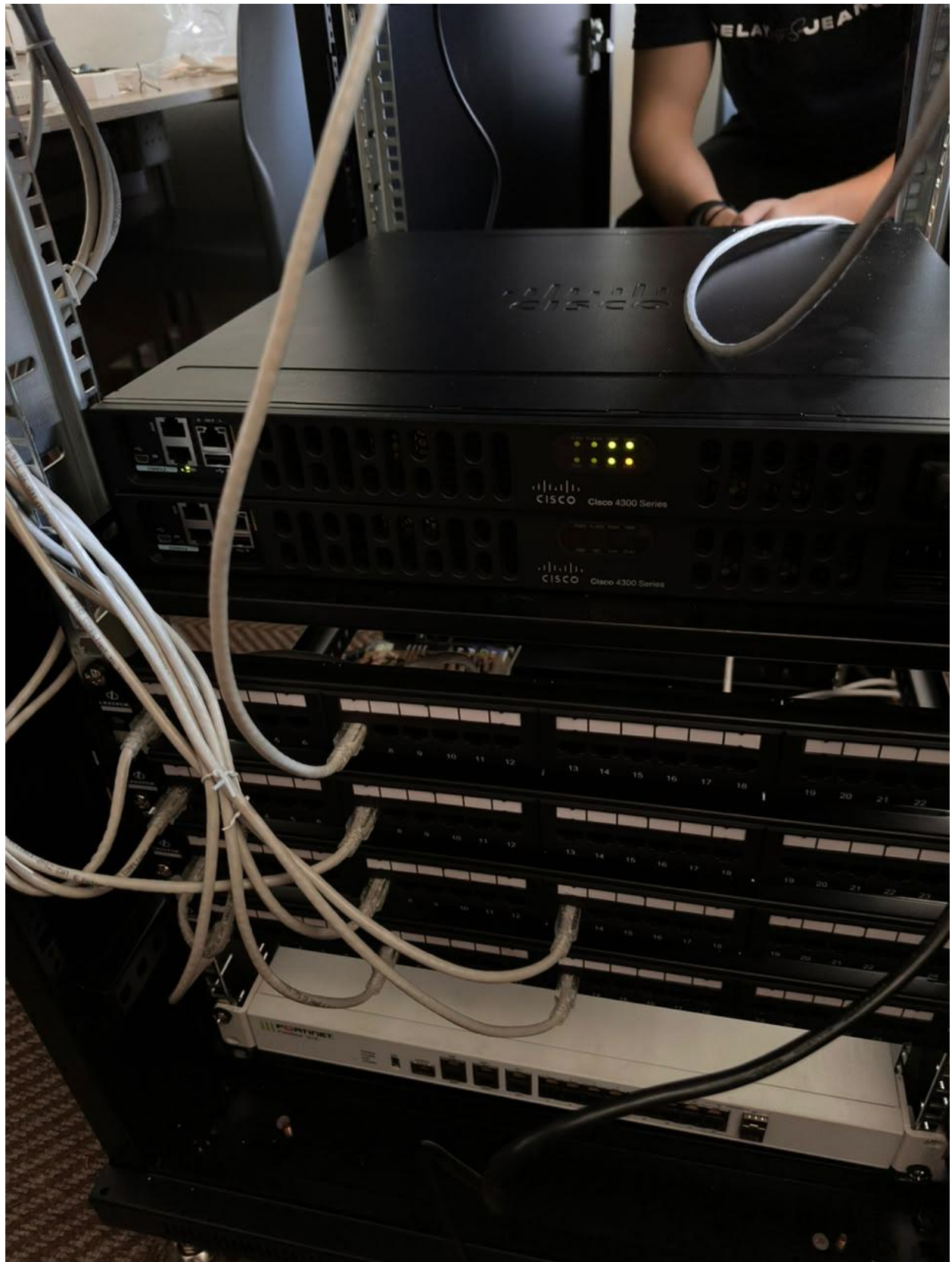
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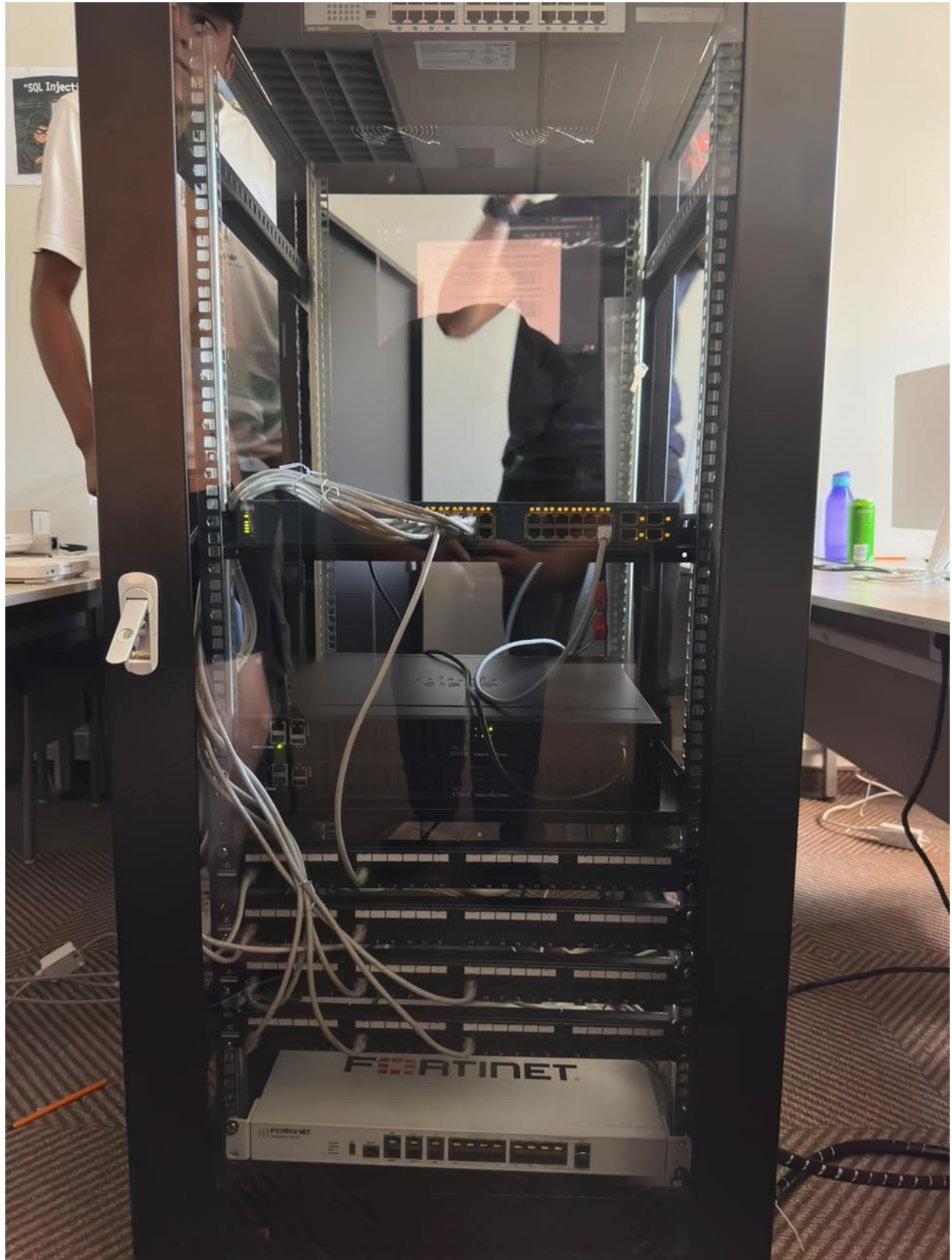
















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PATCH PANEL
SET UP

4	PC 3	PC 5	
3	PC 2	PC 1	
2	PC 9	PC 6	PC 4
1	PC 7	PC 10	PC 8





